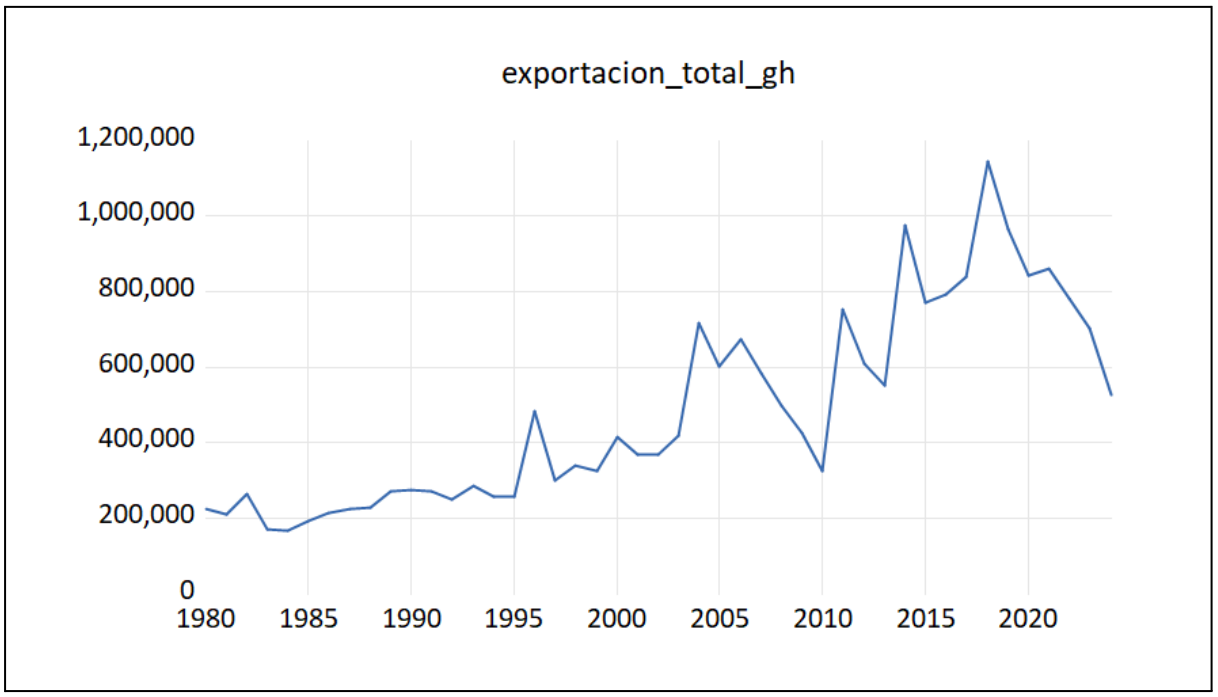
Para este trabajo voy a tratar de explicar las exportaciones de cacao de Ghana, durante el periodo 1980-2024, en función de tres variables explicativas:

- **El precio internacional del cacao**
- **El precio de productor** que establece el gobierno ghanes antes de la cosecha de cada año
- Y el **Incentivo de Contrabando**, que representa el ratio entre el precio de productor de Ghana con respecto al de Costa de Marfil, país vecino y que ambos producen el 60% del cacao a nivel mundial, por lo que un incentivo al contrabando mayor a 1, representa que el precio de Ghana es mayor al de Costa de Marfil, y por lo tanto debería aumentar la exportación en Ghana; si por el contrario es menor a 1, existe un incentivo del productor ghanes a contrabandear cacao hacia Costa de Marfil para obtener un mejor precio, y en consecuencia se vería reducida la exportación oficial de cacao de Ghana.

Por lo tanto necesito trabajar la estacionariedad de las cuatro series y determinar su orden de integración, evaluando la presencia de tendencias determinísticas o estocásticas, evaluar la presencia de raíces unitarias mediante el Test de Dickey Fuller, incorporar variables de intervención ante la presencia de shock aleatorios y se implementaran componentes AR/MA a traves del análisis de las funciones de autocorrelación y autocorrelaciones parciales para llegar a un modelo con residuos ruido blanco, y evaluar la significatividad de los coeficientes (α=5%), que las raíces cumplan las condiciones de estacionariedad e invertibilidad, y analizar la normalidad de los residuos mediante el test de Jarque Bera. Por último cabe aclarar que utilizaré el modelo de máxima verosimilitud (ML) para evitar problemas por falta de convergencia en los modelos con componentes de medias móviles (MA).



El gráfico crece muy lentamente entre 1980-2000, luego se ve una subida con una gran volatilidad, con dos grandes caídas entre 2006-2010 y 2018-2024.

Analizo los test de Dickey Fuller para verificar la existencia de raíz unitaria, empezando con el modelo con tendencia determinística e intercepto:

Null Hypothesis: EXPORTACION TOTAL GH has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.756365	0.0287
Test critical values:	1% level	-4.180911	
	5% level	-3.515523	
	10% level	-3.188259	

*MacKinnon (1996) one-sided p-values.

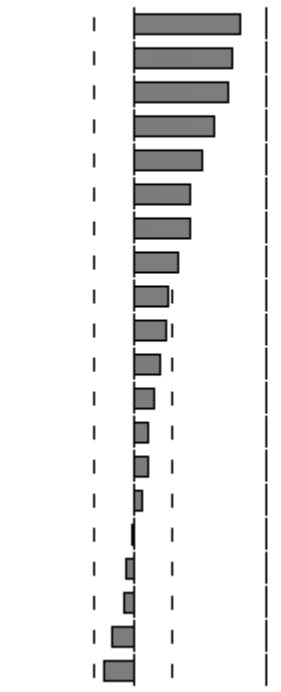
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(EXPORTACION TOTAL GH)
Method: Least Squares
Date: 10/14/26 Time: 18:31
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EXPORTACION TOTAL GH(-1)	-0.582623	0.155103	-3.756365	0.0005
C	64884.70	40058.45	1.619751	0.1130
@TREND("1980")	9894.994	3182.913	3.108785	0.0034

R-squared	0.258826	Mean dependent var	6913.498
Adjusted R-squared	0.222671	S.D. dependent var	142028.8
S.E. of regression	125221.5	Akaike info criterion	26.37930
Sum squared resid	6.43E+11	Schwarz criterion	26.50095
Log likelihood	-577.3446	Hannan-Quinn criter.	26.42441
F-statistic	7.158810	Durbin-Watson stat	1.952845
Prob(F-statistic)	0.002155		

Como la tendencia determinística resulta significativo analizo el p-valor para verificar la presencia de raíz unitaria al 5% de significatividad, como el p-valor es 0.0287<0.05, Rechazo la hipótesis nula de raíz unitaria y por lo tanto analizo la serie en niveles.

Date: 10/14/26 Time: 18:32
Sample: 1980 2024
Included observations: 45



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.843	0.843	34.164	0.000
		2	0.778	0.232	63.913	0.000
		3	0.739	0.147	91.433	0.000
		4	0.646	-0.139	112.99	0.000
		5	0.540	-0.183	128.43	0.000
		6	0.455	-0.088	139.67	0.000
		7	0.446	0.261	150.74	0.000
		8	0.360	-0.083	158.16	0.000
		9	0.277	-0.124	162.65	0.000
		10	0.258	0.029	166.69	0.000
		11	0.209	-0.026	169.41	0.000
		12	0.164	0.053	171.13	0.000
		13	0.120	-0.017	172.08	0.000
		14	0.110	-0.007	172.91	0.000
		15	0.069	-0.092	173.25	0.000
		16	-0.006	-0.138	173.25	0.000
		17	-0.053	-0.117	173.46	0.000
		18	-0.081	0.074	173.98	0.000
		19	-0.166	-0.148	176.23	0.000
		20	-0.223	-0.035	180.45	0.000

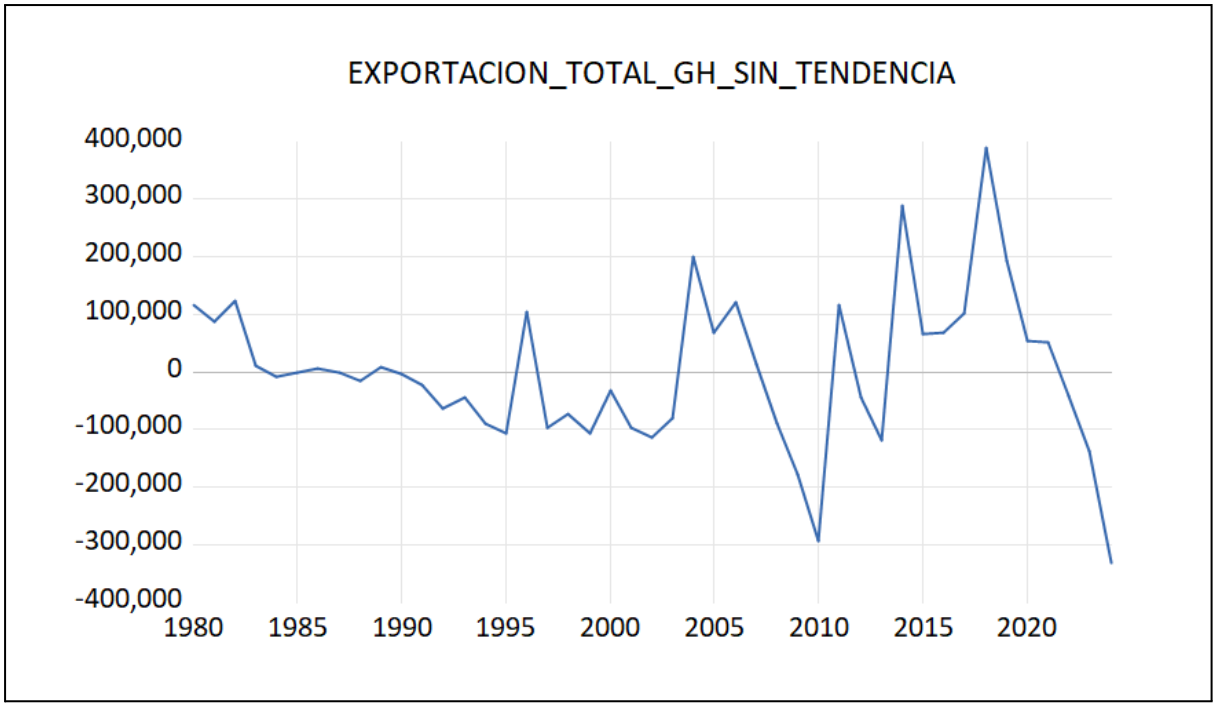
Compruebo la significatividad de la tendencia:

Dependent Variable: EXPORTACION TOTAL GH
Method: Least Squares
Date: 10/14/26 Time: 18:37
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	106537.9	39278.73	2.712355	0.0096
@TREND	17096.48	1537.488	11.11975	0.0000

R-squared	0.741972	Mean dependent var	482660.4
Adjusted R-squared	0.735972	S.D. dependent var	260679.8
S.E. of regression	133946.9	Akaike info criterion	26.49170
Sum squared resid	7.71E+11	Schwarz criterion	26.57200
Log likelihood	-594.0633	Hannan-Quinn criter.	26.52163
F-statistic	123.6488	Durbin-Watson stat	1.130227
Prob(F-statistic)	0.000000		

Como tanto la tendencia como el intercepto resultan significativos, hago la serie de los residuos para evaluar la serie sin tendencia:



Date: 10/14/26 Time: 18:39
Sample: 1980 2024
Included observations: 45

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.355	0.355	6.0601	0.014
		2	0.168	0.049	7.4566	0.024
		3	0.110	0.041	8.0679	0.045
		4	-0.058	-0.133	8.2419	0.083
		5	-0.267	-0.256	12.024	0.034
		6	-0.329	-0.195	17.903	0.006
		7	0.002	0.278	17.903	0.012
		8	-0.100	-0.094	18.477	0.018
		9	-0.205	-0.224	20.935	0.013
		10	-0.072	-0.082	21.246	0.019
		11	0.059	0.090	21.464	0.029
		12	0.011	0.048	21.473	0.044
		13	0.018	0.061	21.495	0.064
		14	0.227	0.062	25.022	0.034
		15	0.125	-0.134	26.123	0.037
		16	-0.050	-0.101	26.307	0.050
		17	-0.187	-0.191	28.938	0.035
		18	-0.104	0.015	29.782	0.040
		19	-0.207	-0.053	33.267	0.022
		20	-0.213	0.006	37.100	0.011

Se ve como cambia el correlograma con respecto al anterior, tanto la FAC como la FACP dejan de ser significativas en el segundo rezago

Dependent Variable: EXPORTACION TOTAL GH SIN TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 14:33
Sample: 1980 2024
Included observations: 45
Convergence achieved after 17 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.538753	0.485760	1.109093	0.2737
MA(1)	-0.148726	0.598438	-0.248524	0.8049
SIGMASQ	1.45E+10	2.27E+09	6.387427	0.0000

R-squared	0.153459	Mean dependent var	1.24E-10
Adjusted R-squared	0.113148	S.D. dependent var	132416.0
S.E. of regression	124699.9	Akaike info criterion	26.37395
Sum squared resid	6.53E+11	Schwarz criterion	26.49440
Log likelihood	-590.4139	Hannan-Quinn criter.	26.41885
Durbin-Watson stat	1.868217		

Inverted AR Roots	.54
Inverted MA Roots	.15

Date: 11/05/26 Time: 14:34
Sample: 1980 2024
Q-statistic probabilities adjusted for 2 ARMA terms

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	-0.001	-0.001	3.E-05	
		2	0.000	0.000	4.E-05	
		3	0.111	0.111	0.6163	0.432
		4	-0.005	-0.005	0.6178	0.734
		5	-0.195	-0.198	2.6363	0.451
		6	-0.329	-0.360	8.5122	0.075
		7	0.146	0.153	9.6948	0.084
		8	-0.048	0.028	9.8280	0.132
		9	-0.217	-0.176	12.605	0.082
		10	-0.024	-0.149	12.640	0.125
		11	0.045	-0.082	12.764	0.174
		12	-0.018	-0.020	12.785	0.236
		13	-0.073	0.032	13.141	0.284
		14	0.202	0.111	15.914	0.195
		15	0.143	0.004	17.361	0.183
		16	-0.014	-0.031	17.375	0.237
		17	-0.155	-0.267	19.188	0.205
		18	0.058	-0.030	19.451	0.246
		19	-0.145	-0.113	21.153	0.220
		20	-0.169	-0.008	23.579	0.169

Dependent Variable: EXPORTACION TOTAL GH SIN TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 14:35
Sample: 1980 2024
Included observations: 45
Convergence achieved after 16 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.420124	0.274241	1.531953	0.1332
MA(1)	-0.058044	0.260608	-0.222725	0.8249
MA(6)	-0.359507	0.148547	-2.420165	0.0200
SIGMASQ	1.26E+10	2.44E+09	5.167792	0.0000

R-squared	0.265614	Mean dependent var	1.24E-10
Adjusted R-squared	0.211879	S.D. dependent var	132416.0
S.E. of regression	117553.9	Akaike info criterion	26.29361
Sum squared resid	5.67E+11	Schwarz criterion	26.45421
Log likelihood	-587.6063	Hannan-Quinn criter.	26.35348
Durbin-Watson stat	1.983371		

Inverted AR Roots	.42			
Inverted MA Roots	.85	.43-.73i	.43+.73i	-.41-.73i
	-.41+.73i	-.83		

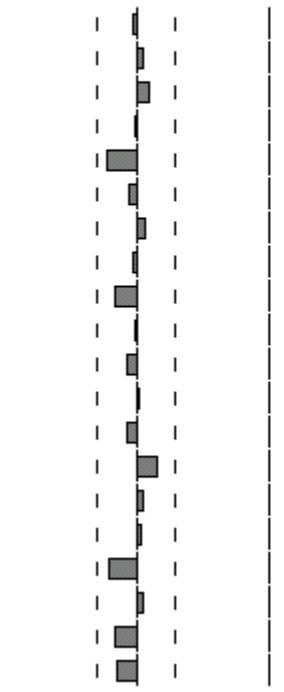
Dependent Variable: EXPORTACION TOTAL GH SIN TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 14:36
Sample: 1980 2024
Included observations: 45
Convergence achieved after 12 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.369337	0.124035	2.977689	0.0048
MA(6)	-0.353814	0.148475	-2.382993	0.0218
SIGMASQ	1.26E+10	2.32E+09	5.432109	0.0000

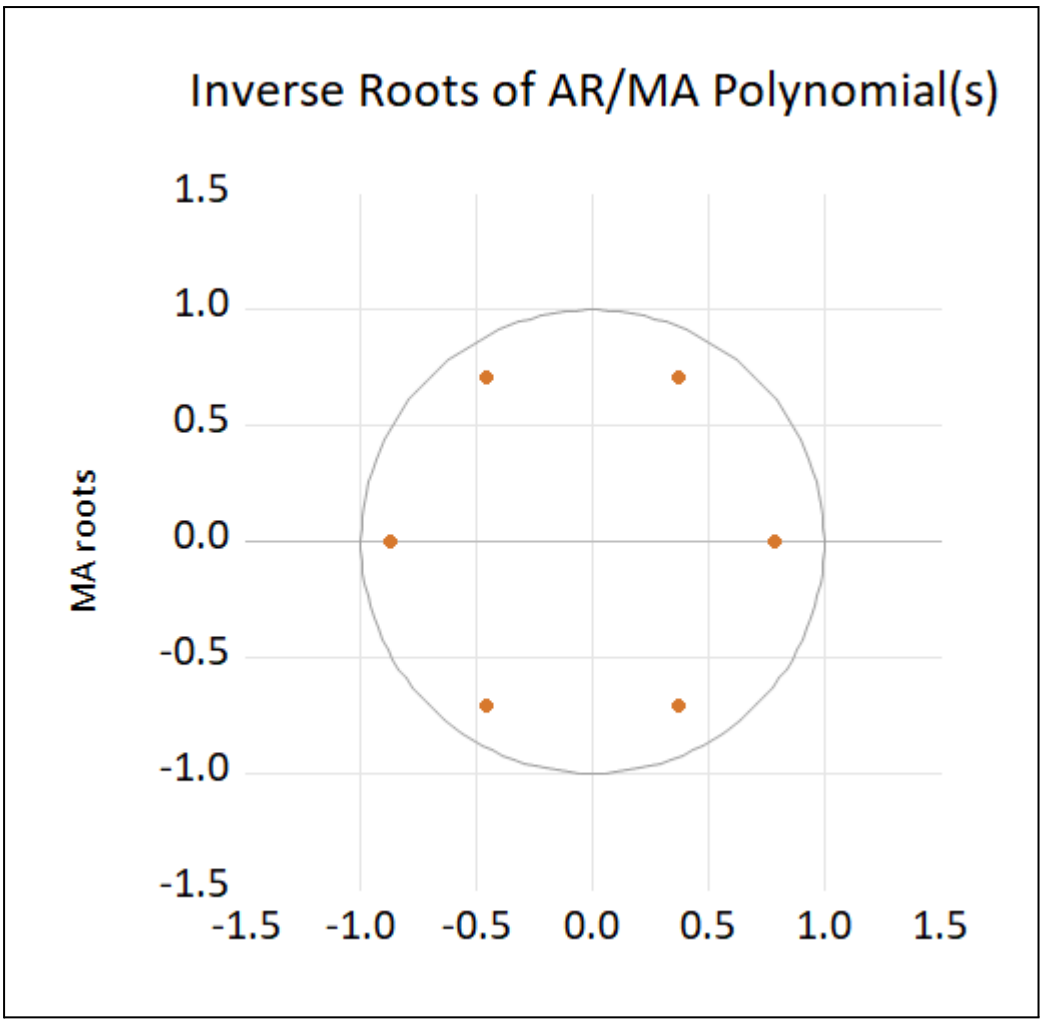
R-squared	0.264286	Mean dependent var	1.24E-10
Adjusted R-squared	0.229252	S.D. dependent var	132416.0
S.E. of regression	116251.0	Akaike info criterion	26.25028
Sum squared resid	5.68E+11	Schwarz criterion	26.37073
Log likelihood	-587.6314	Hannan-Quinn criter.	26.29518
Durbin-Watson stat	1.991625		

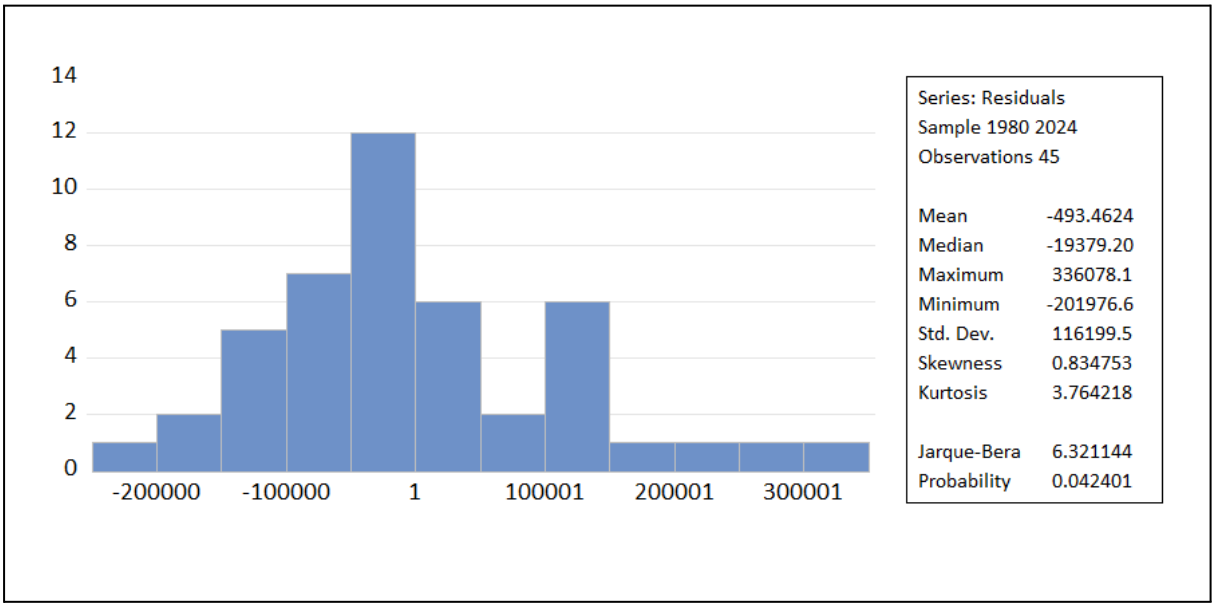
Inverted AR Roots	.37			
Inverted MA Roots	.84	.42+.73i	.42-.73i	-.42+.73i
	-.42-.73i	-.84		

Date: 11/05/26 Time: 14:36
Sample: 1980 2024
Q-statistic probabilities adjusted for 2 ARMA terms

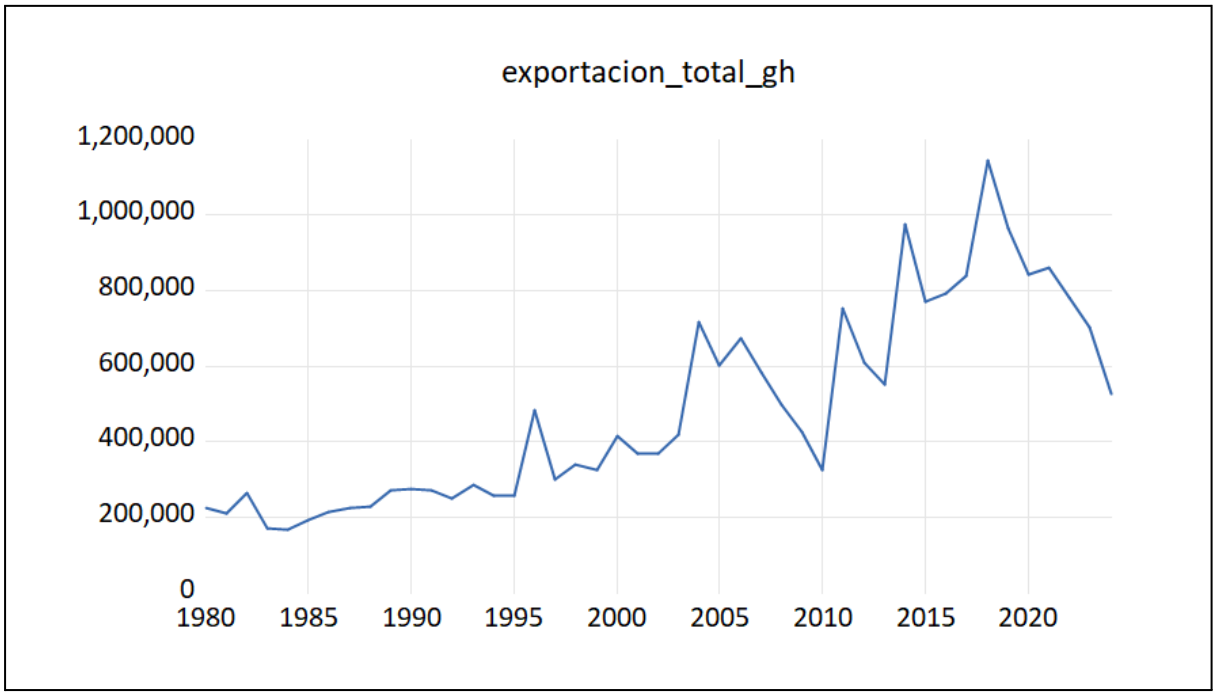


	AC	PAC	Q-Stat	Prob
1	-0.028	-0.028	0.0387	
2	0.051	0.051	0.1684	
3	0.103	0.106	0.7030	0.402
4	-0.011	-0.008	0.7095	0.701
5	-0.237	-0.252	3.6771	0.299
6	-0.055	-0.087	3.8385	0.428
7	0.076	0.114	4.1595	0.527
8	-0.017	0.064	4.1766	0.653
9	-0.162	-0.186	5.7267	0.572
10	-0.010	-0.135	5.7333	0.677
11	-0.071	-0.086	6.0513	0.735
12	0.018	0.137	6.0717	0.809
13	-0.080	-0.030	6.4986	0.838
14	0.167	0.066	8.3924	0.754
15	0.062	-0.003	8.6609	0.798
16	0.036	0.032	8.7549	0.846
17	-0.206	-0.247	11.965	0.682
18	0.051	-0.017	12.171	0.732
19	-0.167	-0.115	14.451	0.635
20	-0.153	-0.087	16.420	0.563


Inverse Roots of AR/MA Polynomial(s)
MA roots
1.5
1.0
0.5
0.0
-0.5
-1.0
-1.5
-1.5 -1.0 -0.5 0.0 0.5 1.0 1.5


14
12
10
8
6
4
2
0
-200000 -100000 1 100001 200001 300001
Series: Residuals
Sample 1980 2024
Observations 45
Mean -493.4624
Median -19379.20
Maximum 336078.1
Minimum -201976.6
Std. Dev. 116199.5
Skewness 0.834753
Kurtosis 3.764218
Jarque-Bera 6.321144
Probability 0.042401

Incorporando una variable de impacto:



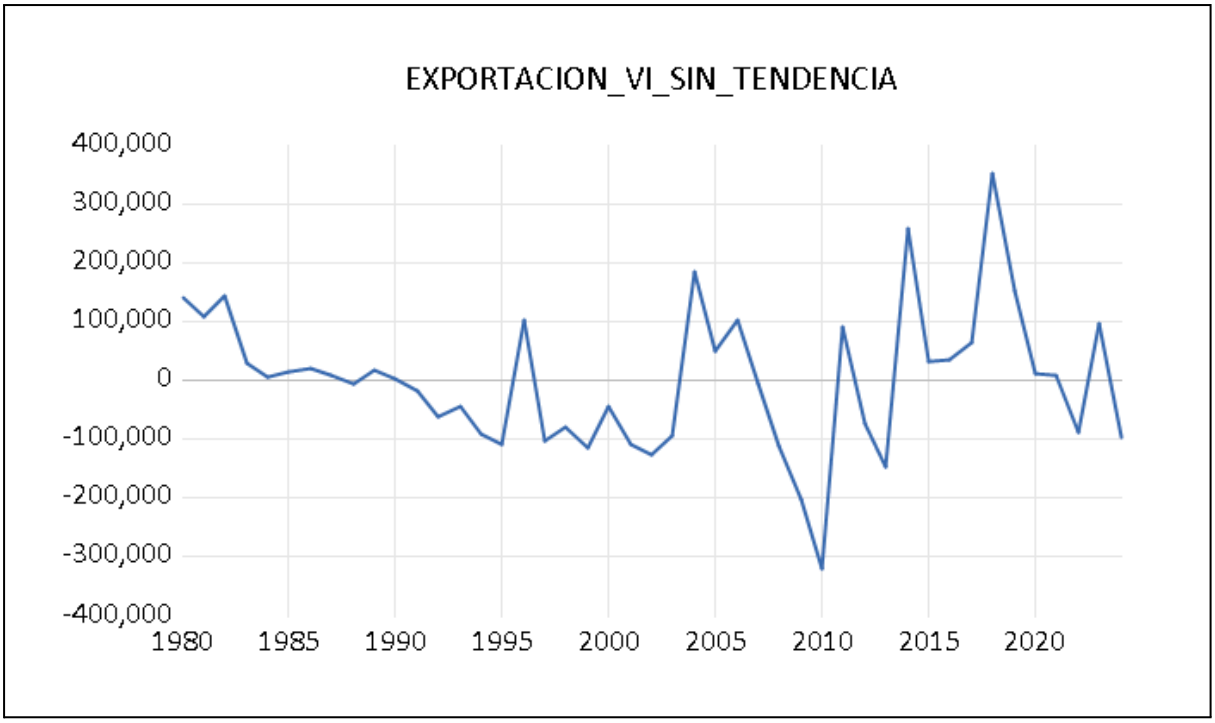
Durante los últimos años está ocurriendo una crisis respecto a la producción del cacao en África Occidental, provocada por un virus sobre la planta de cacao, que afectó a Ghana y Costa de Marfil, dos de los mayores productores de cacao a nivel mundial (60% de la oferta mundial).
Es por eso que decidí incorporar una variable de intervención que refleje ese shock tanto en la producción como en el precio internacional, y por lo tanto en el precio del productor.
Con respecto a la variable exportación total

Dependent Variable: EXPORTACION_TOTAL_GH
Method: Least Squares
Date: 11/02/26 Time: 12:34
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
VI_CRISIS	-281198.5	95567.65	-2.942403	0.0053
C	83987.63	36990.97	2.270490	0.0284
@TREND	18689.57	1516.467	12.32441	0.0000

R-squared	0.786071	Mean dependent var	482660.4
Adjusted R-squared	0.775884	S.D. dependent var	260679.8
S.E. of regression	123408.2	Akaike info criterion	26.34872
Sum squared resid	6.40E+11	Schwarz criterion	26.46917
Log likelihood	-589.8463	Hannan-Quinn criter.	26.39362
F-statistic	77.16340	Durbin-Watson stat	1.403986
Prob(F-statistic)	0.000000		

La variable de intervención que mejor refleja la crisis en África occidental es aquella que incluye a los años 2023 y 2024.



Null Hypothesis: EXPORTACION_VI_SIN_TENDENCIA has a unit root
Exogenous: None
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

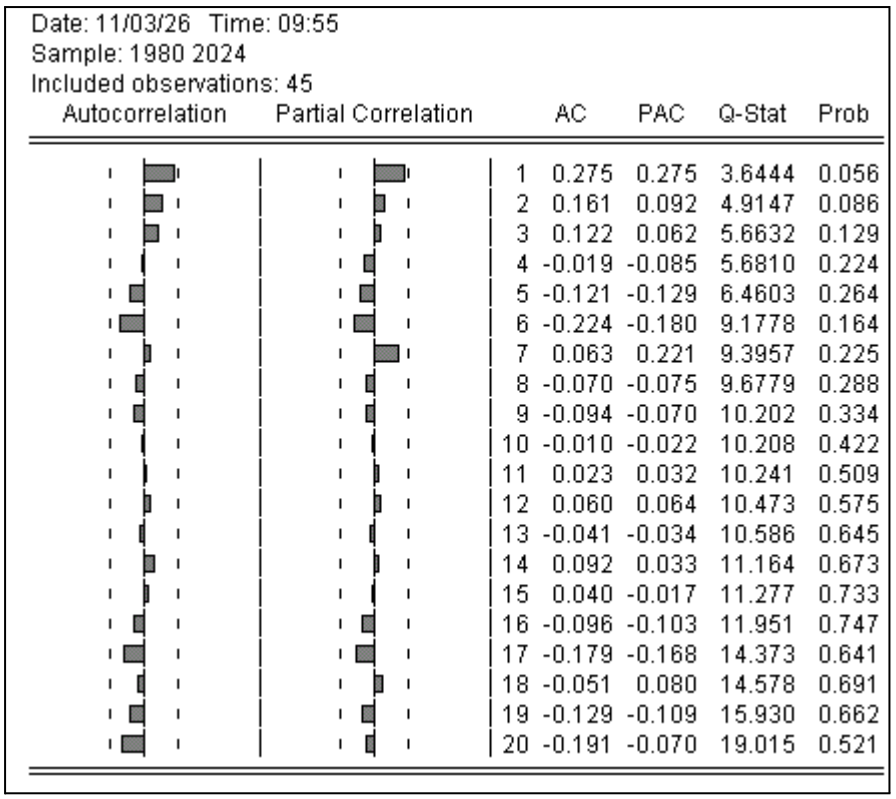
		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-4.964373	0.0000
Test critical values:	1% level	-2.618579	
	5% level	-1.948495	
	10% level	-1.612135	

*MacKinnon (1996) one-sided p-values.

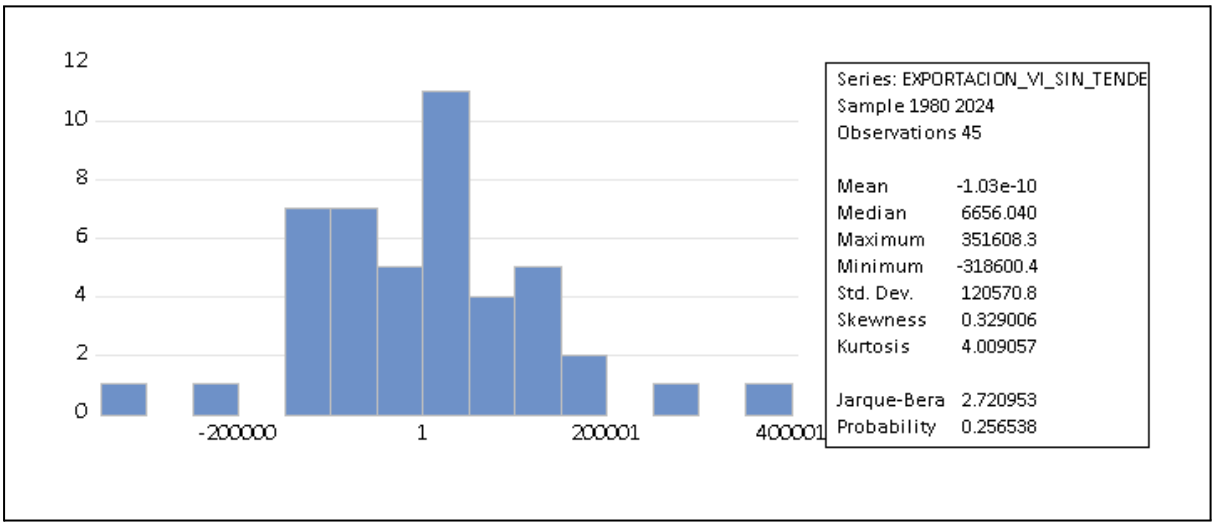
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(EXPORTACION_VI_SIN_TENDENCIA)
Method: Least Squares
Date: 11/02/26 Time: 12:36
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EXPORTACION_VI_SIN_TENDENCIA(-1)	-0.720531	0.145140	-4.964373	0.0000

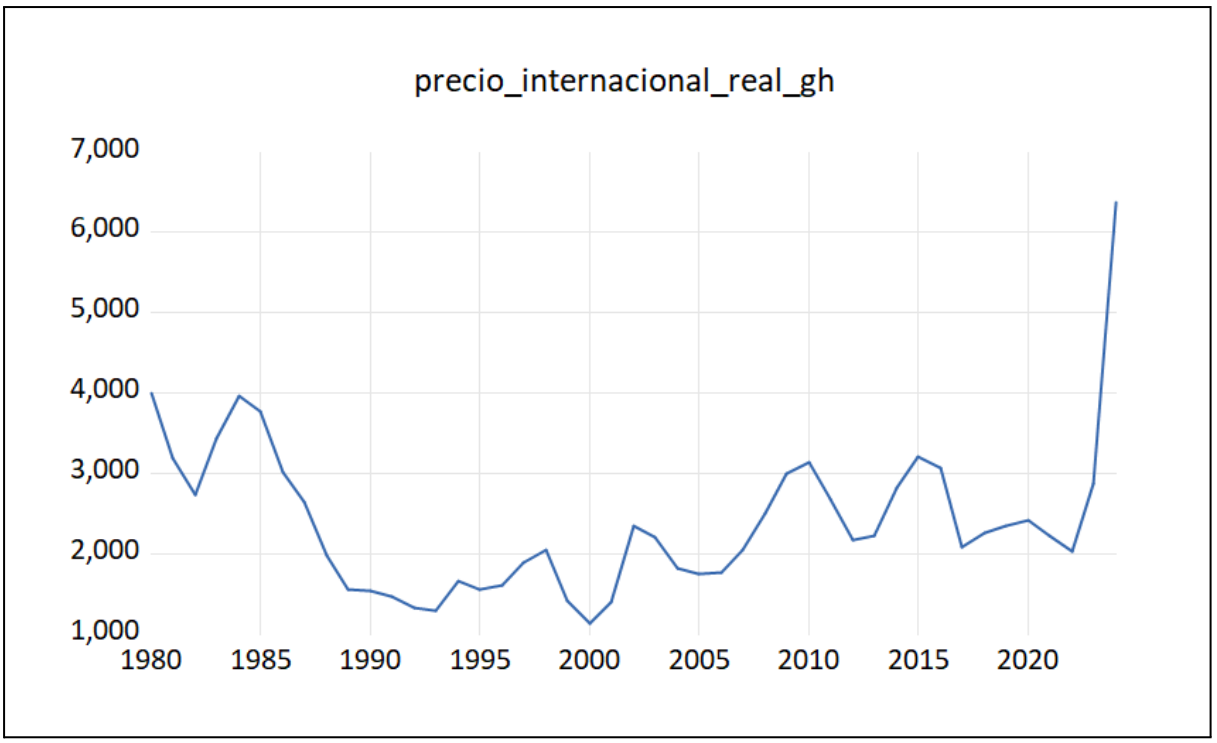
R-squared	0.363424	Mean dependent var	-5385.196
Adjusted R-squared	0.363424	S.D. dependent var	144413.2
S.E. of regression	115221.1	Akaike info criterion	26.16956
Sum squared resid	5.71E+11	Schwarz criterion	26.21011
Log likelihood	-574.7303	Hannan-Quinn criter.	26.18460
Durbin-Watson stat	2.048261		



	AC	PAC	Q-Stat	Prob
1	0.275	0.275	3.6444	0.056
2	0.161	0.092	4.9147	0.086
3	0.122	0.062	5.6632	0.129
4	-0.019	-0.085	5.6810	0.224
5	-0.121	-0.129	6.4603	0.264
6	-0.224	-0.180	9.1778	0.164
7	0.063	0.221	9.3957	0.225
8	-0.070	-0.075	9.6779	0.288
9	-0.094	-0.070	10.202	0.334
10	-0.010	-0.022	10.208	0.422
11	0.023	0.032	10.241	0.509
12	0.060	0.064	10.473	0.575
13	-0.041	-0.034	10.586	0.645
14	0.092	0.033	11.164	0.673
15	0.040	-0.017	11.277	0.733
16	-0.096	-0.103	11.951	0.747
17	-0.179	-0.168	14.373	0.641
18	-0.051	0.080	14.578	0.691
19	-0.129	-0.109	15.930	0.662
20	-0.191	-0.070	19.015	0.521



Al ver el correlograma luego de eliminar la tendencia, el intercepto y el efecto de la crisis de los anios 23/24 arroja directamente ruido blanco, por lo que no hay ningún componente AR/MA que incluir en el modelo.
Además el comportamiento de la nueva serie no rechaza el test de normalidad.



En el gráfico se observa una fuerte caída entre 1985 y 1990, luego cierta estabilidad, y finalmente una gran subida en los últimos años del precio internacional del cacao.

Analizo los test de Dickey Fuller para verificar la existencia de raíz unitaria, empezando con el modelo con tendencia determinística e intercepto:

Null Hypothesis: PRECIO INTERNACIONAL REAL GH has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 1 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-1.243925	0.8882
Test critical values:	1% level	-4.186481	
	5% level	-3.518090	
	10% level	-3.189732	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO INTERNACIONAL REAL GH)
Method: Least Squares
Date: 10/14/26 Time: 18:49
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO INTERNACIONAL REAL GH(-1)	-0.172549	0.138713	-1.243925	0.2210
D(PRECIO INTERNACIONAL REAL GH	0.616396	0.226211	2.724873	0.0096
C	221.7287	385.0076	0.575907	0.5680
@TREND("1980")	11.30600	7.774911	1.454164	0.1539

R-squared	0.240057	Mean dependent var	74.03259
Adjusted R-squared	0.181600	S.D. dependent var	682.1569
S.E. of regression	617.1163	Akaike info criterion	15.77640
Sum squared resid	14852469	Schwarz criterion	15.94023
Log likelihood	-335.1926	Hannan-Quinn criter.	15.83682
F-statistic	4.106556	Durbin-Watson stat	1.119953
Prob(F-statistic)	0.012631		

Como la tendencia determinística no resulta significativa (p-valor = 0.1539), vuelvo a hacer el test con intercepto.

Null Hypothesis: PRECIO INTERNACIONAL REAL GH has a unit root
Exogenous: Constant
Lag Length: 2 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-0.566130	0.8673
Test critical values:	1% level	-3.596616	
	5% level	-2.933158	
	10% level	-2.604867	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO INTERNACIONAL REAL GH)
Method: Least Squares
Date: 10/14/26 Time: 18:49
Sample (adjusted): 1983 2024
Included observations: 42 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO INTERNACIONAL REAL GH(-1)	-0.081703	0.144319	-0.566130	0.5746
D(PRECIO INTERNACIONAL REAL GH	0.834834	0.227768	3.665281	0.0008
D(PRECIO INTERNACIONAL REAL GH	-0.628098	0.235939	-2.662122	0.0113
C	247.0291	340.6667	0.725134	0.4728

R-squared	0.316966	Mean dependent var	86.39314
Adjusted R-squared	0.263043	S.D. dependent var	685.5347
S.E. of regression	588.5057	Akaike info criterion	15.68344
Sum squared resid	13160881	Schwarz criterion	15.84894
Log likelihood	-325.3523	Hannan-Quinn criter.	15.74410
F-statistic	5.878048	Durbin-Watson stat	1.412468
Prob(F-statistic)	0.002123		

Nuevamente, el intercepto no resulta significativo (p-valor = 0.4728), por lo que realizo el test sin tendencia ni intercepto.

Null Hypothesis: PRECIO INTERNACIONAL REAL GH has a unit root
Exogenous: None
Lag Length: 2 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		0.496799	0.8186
Test critical values:	1% level	-2.621185	
	5% level	-1.948886	
	10% level	-1.611932	

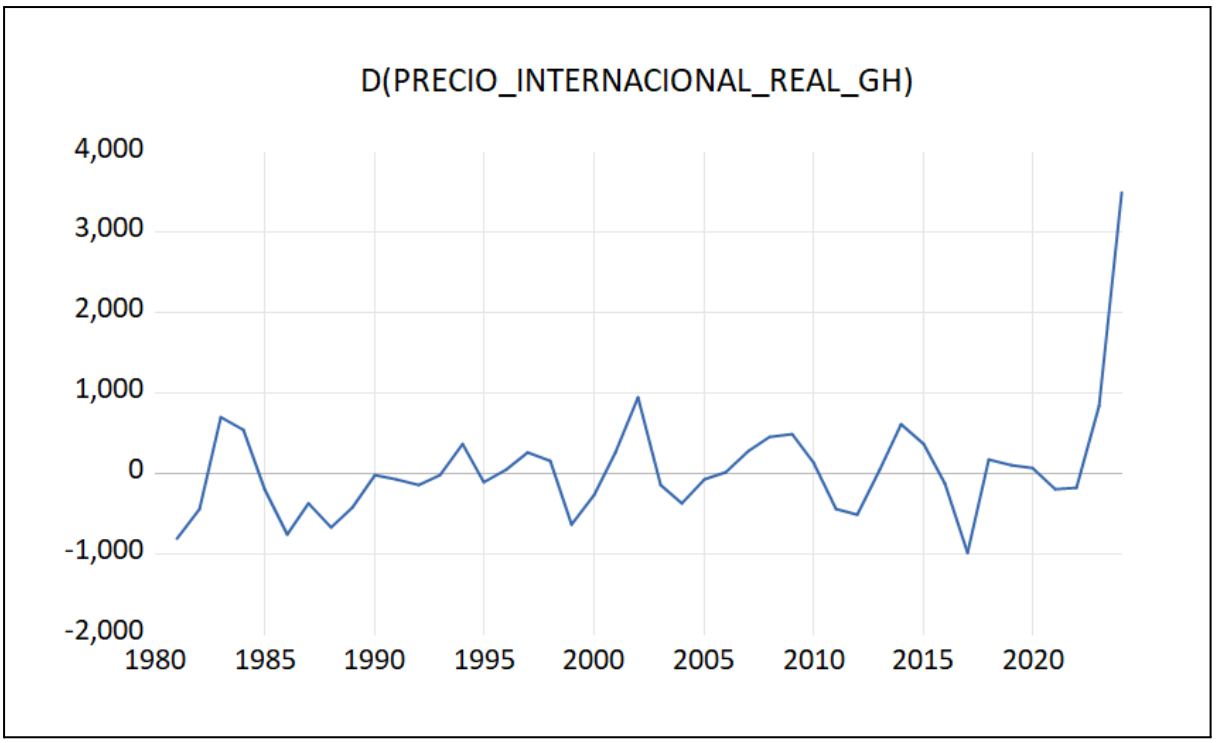
*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO INTERNACIONAL REAL GH)
Method: Least Squares
Date: 10/14/26 Time: 18:50
Sample (adjusted): 1983 2024
Included observations: 42 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO INTERNACIONAL REAL GH(-1)	0.019113	0.038472	0.496799	0.6221
D(PRECIO INTERNACIONAL REAL GH	0.802145	0.221901	3.614878	0.0008
D(PRECIO INTERNACIONAL REAL GH	-0.682925	0.222133	-3.074394	0.0038

R-squared	0.307515	Mean dependent var	86.39314
Adjusted R-squared	0.272003	S.D. dependent var	685.5347
S.E. of regression	584.9171	Akaike info criterion	15.64957
Sum squared resid	13342993	Schwarz criterion	15.77369
Log likelihood	-325.6409	Hannan-Quinn criter.	15.69506
Durbin-Watson stat	1.441096		

Como el p-valor asociado al test estadístico es 0.8186 no rechazo la hipótesis nula de raíz unitaria por lo que debo diferenciar la serie:



Notar la presencia de un valor atípico en los últimos años (2023-2024) donde el precio internacional se dispara.

Realizo el test de Dickey Fuller nuevamente para asegurarme de que no haya que diferenciar nuevamente la serie:

Null Hypothesis: D(PRECIO_INTERNACIONAL_REAL_GH) has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 1 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.690253	0.0341
Test critical values:	1% level	-4.192337	
	5% level	-3.520787	
	10% level	-3.191277	

*MacKinnon (1996) one-sided p-values.

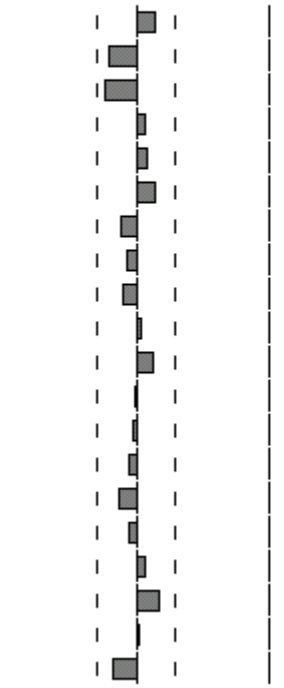
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO_INTERNACIONAL_REAL_GH,2)
Method: Least Squares
Date: 10/20/26 Time: 17:58
Sample (adjusted): 1983 2024
Included observations: 42 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(PRECIO_INTERNACIONAL_REAL_GH(-1	-0.959795	0.260089	-3.690253	0.0007
D(PRECIO_INTERNACIONAL_REAL_GH(-1	0.716317	0.215317	3.326810	0.0020
C	-293.3396	192.6193	-1.522898	0.1361
@TREND("1980")	14.97777	7.259097	2.063311	0.0460

R-squared	0.314819	Mean dependent var	93.76005
Adjusted R-squared	0.260726	S.D. dependent var	651.7977
S.E. of regression	560.4226	Akaike info criterion	15.58565
Sum squared resid	11934794	Schwarz criterion	15.75114
Log likelihood	-323.2987	Hannan-Quinn criter.	15.64631
F-statistic	5.819927	Durbin-Watson stat	1.563827
Prob(F-statistic)	0.002247		

Como la tendencia resulta significativa y el estadístico del test tiene un p-valor = 0.0341, rechazo la hipótesis nula de raíz unitaria en la serie diferenciada, al 5% de significatividad.

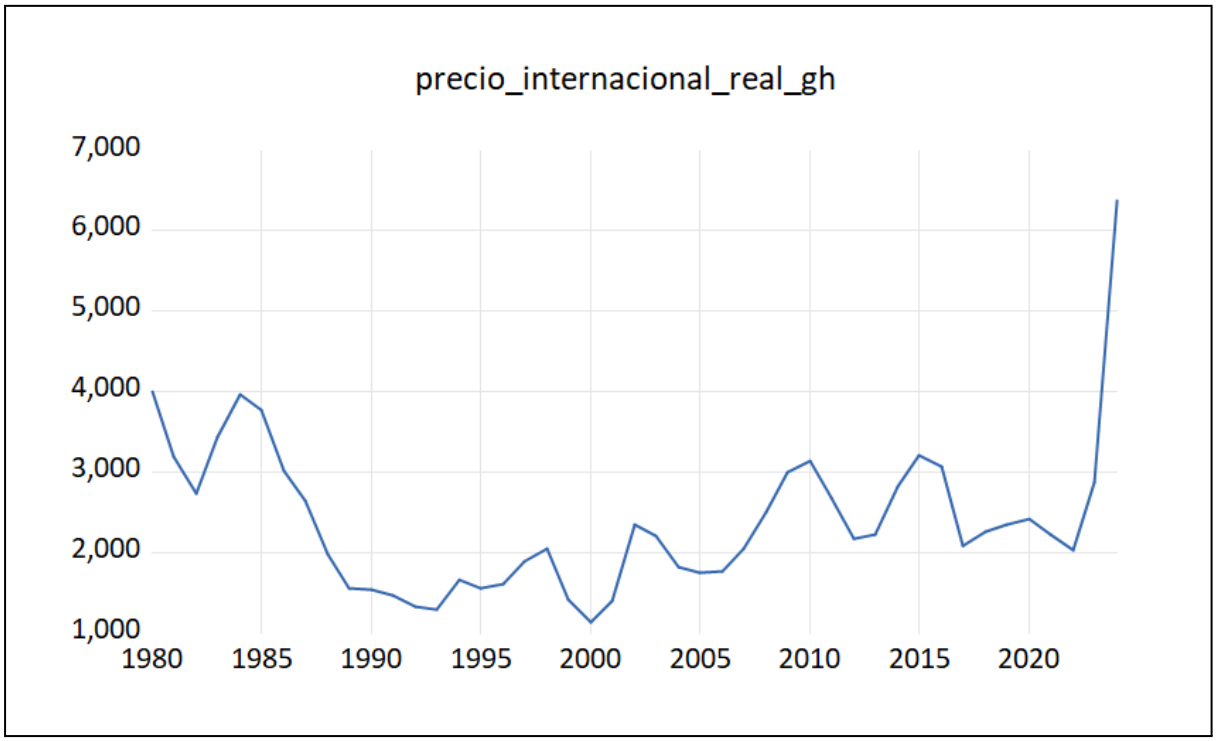
Date: 10/14/26 Time: 18:52
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.145	0.145	0.9644	0.326
		2	-0.219	-0.245	3.2214	0.200
		3	-0.245	-0.185	6.1186	0.106
		4	0.072	0.096	6.3733	0.173
		5	0.084	-0.035	6.7320	0.241
		6	0.148	0.144	7.8796	0.247
		7	-0.110	-0.122	8.5321	0.288
		8	-0.079	0.013	8.8812	0.352
		9	-0.099	-0.091	9.4395	0.398
		10	0.037	-0.017	9.5193	0.484
		11	0.133	0.120	10.582	0.479
		12	-0.000	-0.103	10.582	0.565
		13	-0.017	0.120	10.601	0.644
		14	-0.056	-0.072	10.807	0.701
		15	-0.137	-0.147	12.110	0.671
		16	-0.054	-0.016	12.319	0.722
		17	0.082	-0.035	12.816	0.748
		18	0.181	0.197	15.354	0.638
		19	0.019	-0.059	15.385	0.698
		20	-0.186	-0.075	18.284	0.569

El correlograma de la serie diferenciada se comporta como ruido blanco, lo cual tiene sentido teniendo en cuenta que se trata de la serie de precio internacional del cacao, que al ser una commodity es normal que se comporte como un *random walk*.

Incorporando una variable de intervención:

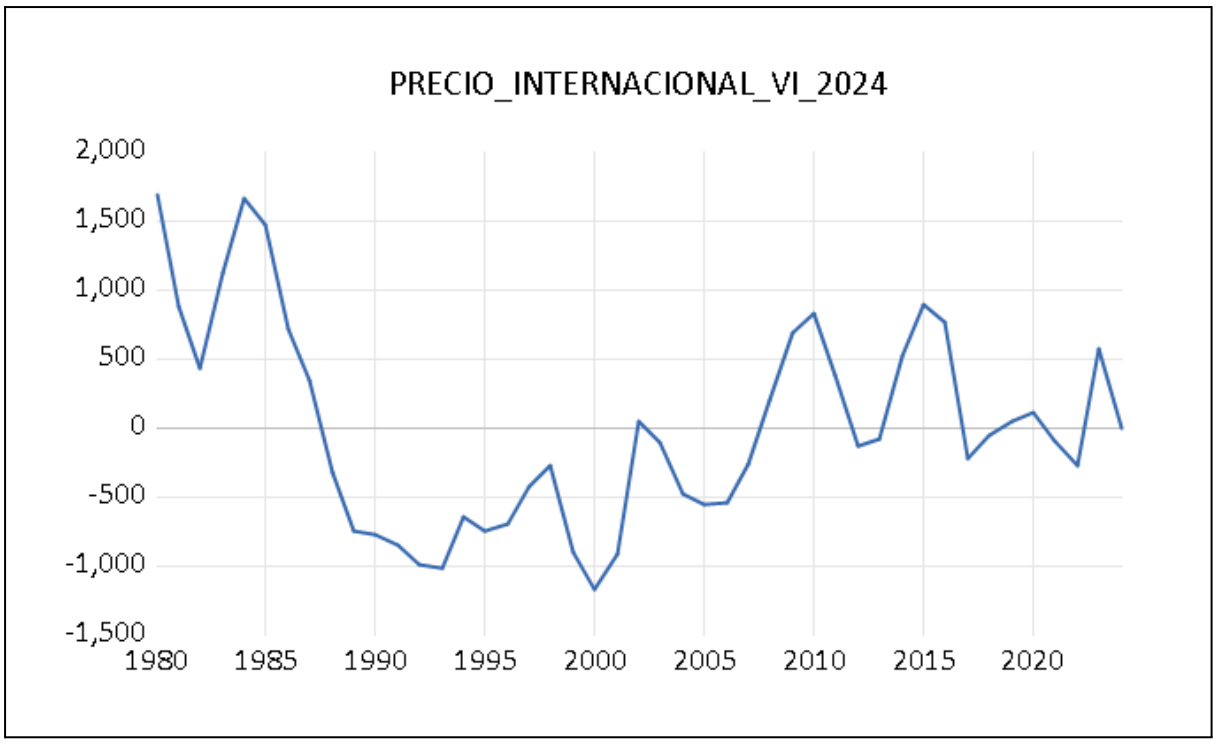


A pesar de que el modelo anterior reportaba un random walk después de identificar una raíz unitaria, lo que tiene sentido económico teniendo en cuenta de que se trata de una commoditie, incorporo una variable de intervención debido al pico en el precio internacional en el año 2024, debido a la escasez de producción de justamente, Ghana y Costa de Marfil, que representa un shock exógeno y transitorio, ya que en los años posteriores, 2025 y durante 2026, se reportó un descenso del precio internacional por la recuperación de la producción de estos países.

Dependent Variable: PRECIO_INTERNACIONAL_REAL_GH
Method: Least Squares
Date: 11/02/26 Time: 12:49
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	2309.389	111.9929	20.62085	0.0000
VI_2024	4058.679	751.2712	5.402415	0.0000

R-squared	0.404317	Mean dependent var	2399.582
Adjusted R-squared	0.390464	S.D. dependent var	951.5185
S.E. of regression	742.8769	Akaike info criterion	16.10236
Sum squared resid	23730240	Schwarz criterion	16.18266
Log likelihood	-360.3032	Hannan-Quinn criter.	16.13230
F-statistic	29.18608	Durbin-Watson stat	0.360362
Prob(F-statistic)	0.000003		



Null Hypothesis: PRECIO_INTERNACIOAL_VI_2024 has a unit root
Exogenous: None
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-2.857434	0.0053
Test critical values:	1% level	-2.618579	
	5% level	-1.948495	
	10% level	-1.612135	

*MacKinnon (1996) one-sided p-values.

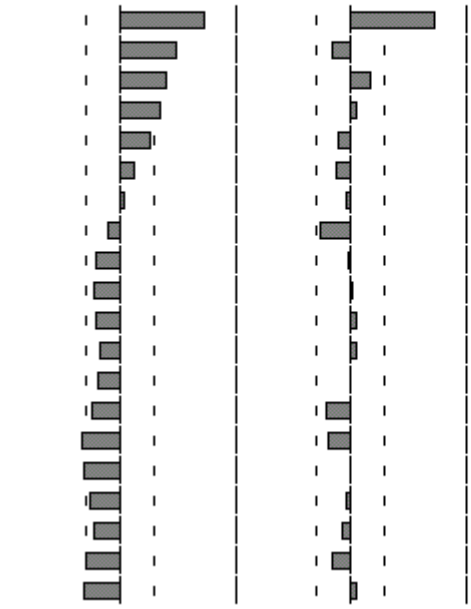
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO_INTERNACIOAL_VI_2024)
Method: Least Squares
Date: 11/02/26 Time: 13:00
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO_INTERNACIOAL_VI_2024(-1)	-0.239806	0.083923	-2.857434	0.0066

R-squared	0.153212	Mean dependent var	-38.23194
Adjusted R-squared	0.153212	S.D. dependent var	444.2701
S.E. of regression	408.8221	Akaike info criterion	14.88690
Sum squared resid	7186828.	Schwarz criterion	14.92745
Log likelihood	-326.5119	Hannan-Quinn criter.	14.90194
Durbin-Watson stat	1.412971		

Ahora el test de Dickey Fuller rechaza la hipótesis nula de raíz unitaria, al contrario que el modelo anterior sin variable de intervención.

Date: 11/02/26 Time: 13:01
Sample: 1980 2024
Included observations: 45



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.760	0.760	27.778	0.000
		2	0.508	-0.165	40.483	0.000
		3	0.402	0.188	48.620	0.000
		4	0.362	0.047	55.361	0.000
		5	0.270	-0.104	59.204	0.000
		6	0.129	-0.119	60.106	0.000
		7	0.028	-0.030	60.149	0.000
		8	-0.113	-0.270	60.872	0.000
		9	-0.215	-0.018	63.592	0.000
		10	-0.240	0.014	67.076	0.000
		11	-0.211	0.053	69.833	0.000
		12	-0.183	0.049	71.970	0.000
		13	-0.189	-0.004	74.343	0.000
		14	-0.254	-0.216	78.742	0.000
		15	-0.344	-0.197	87.068	0.000
		16	-0.336	-0.000	95.286	0.000
		17	-0.265	-0.026	100.59	0.000
		18	-0.240	-0.069	105.09	0.000
		19	-0.314	-0.161	113.12	0.000
		20	-0.332	0.056	122.42	0.000

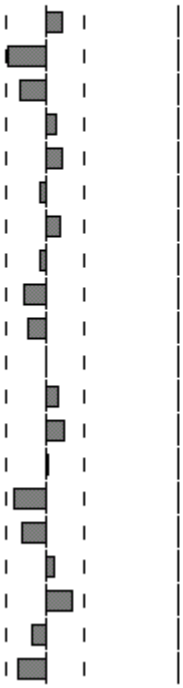
Dependent Variable: PRECIO INTERNACIONAL VI 2024
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 12:19
Sample: 1980 2024
Included observations: 45
Convergence achieved after 18 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.838570	0.066968	12.52203	0.0000
SIGMASQ	181610.8	44624.54	4.069752	0.0002

R-squared	0.655609	Mean dependent var	5.00E-14
Adjusted R-squared	0.647600	S.D. dependent var	734.3866
S.E. of regression	435.9562	Akaike info criterion	15.06338
Sum squared resid	8172487.	Schwarz criterion	15.14368
Log likelihood	-336.9261	Hannan-Quinn criter.	15.09331
Durbin-Watson stat	1.595108		

Inverted AR Roots	.84

Date: 11/05/26 Time: 12:20
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.135	0.135	0.8754	
		2	-0.291	-0.315	5.0526	0.025
		3	-0.204	-0.124	7.1513	0.028
		4	0.082	0.048	7.5001	0.058
		5	0.127	0.016	8.3583	0.079
		6	-0.043	-0.061	8.4577	0.133
		7	0.116	0.220	9.2130	0.162
		8	-0.045	-0.123	9.3275	0.230
		9	-0.162	-0.098	10.879	0.209
		10	-0.139	-0.094	12.041	0.211
		11	0.005	-0.080	12.042	0.282
		12	0.095	-0.032	12.623	0.319
		13	0.143	0.178	13.976	0.302
		14	0.023	-0.013	14.013	0.373
		15	-0.245	-0.179	18.229	0.197
		16	-0.174	-0.051	20.427	0.156
		17	0.075	-0.011	20.853	0.184
		18	0.214	0.032	24.424	0.108
		19	-0.102	-0.172	25.269	0.118
		20	-0.212	-0.124	29.065	0.065

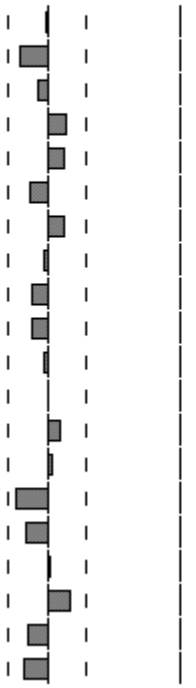
Dependent Variable: PRECIO INTERNACIONAL VI 2024
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 12:20
Sample: 1980 2024
Included observations: 45
Convergence achieved after 8 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	1.100144	0.139435	7.890003	0.0000
AR(2)	-0.319626	0.143193	-2.232128	0.0310
SIGMASQ	164817.5	37039.64	4.449760	0.0001

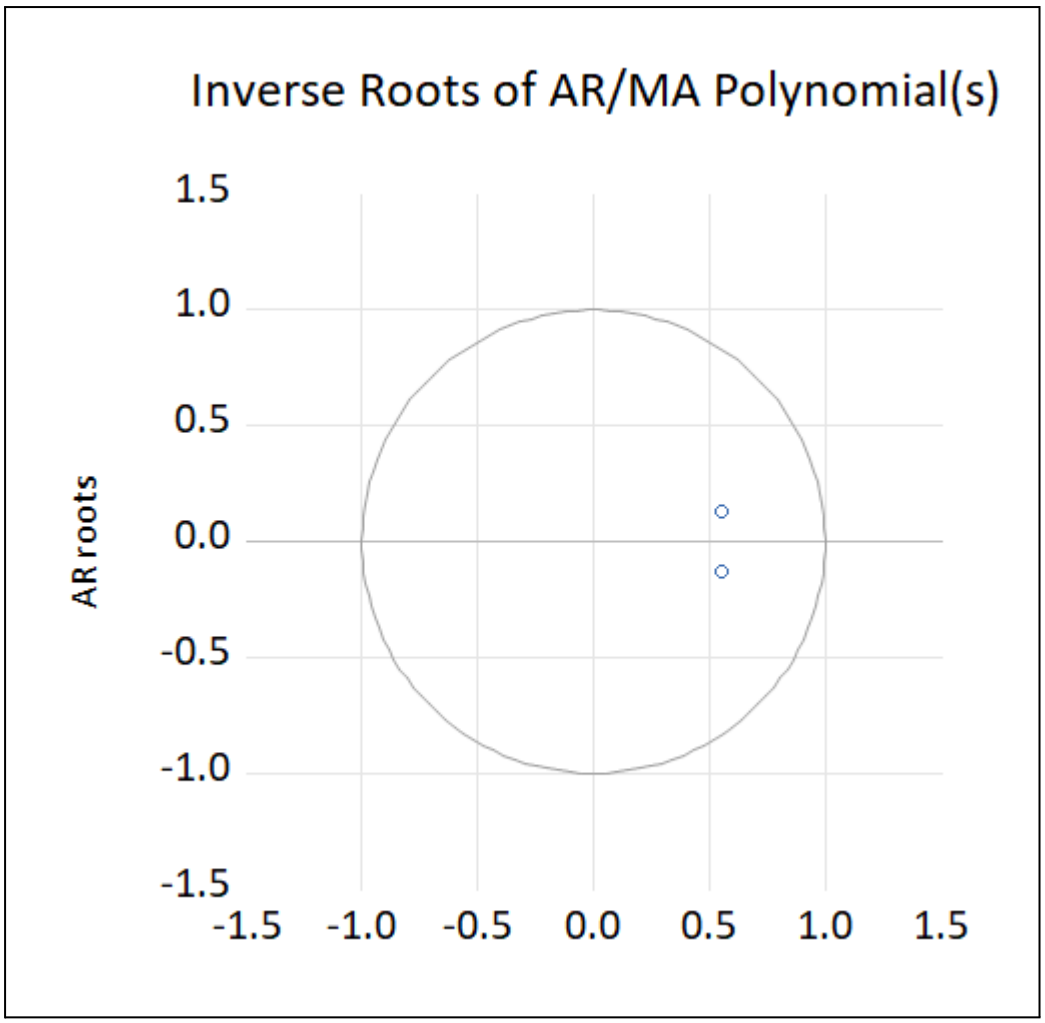
R-squared	0.687454	Mean dependent var	5.00E-14
Adjusted R-squared	0.672571	S.D. dependent var	734.3866
S.E. of regression	420.2264	Akaike info criterion	15.01498
Sum squared resid	7416789.	Schwarz criterion	15.13543
Log likelihood	-334.8371	Hannan-Quinn criter.	15.05988
Durbin-Watson stat	1.853960		

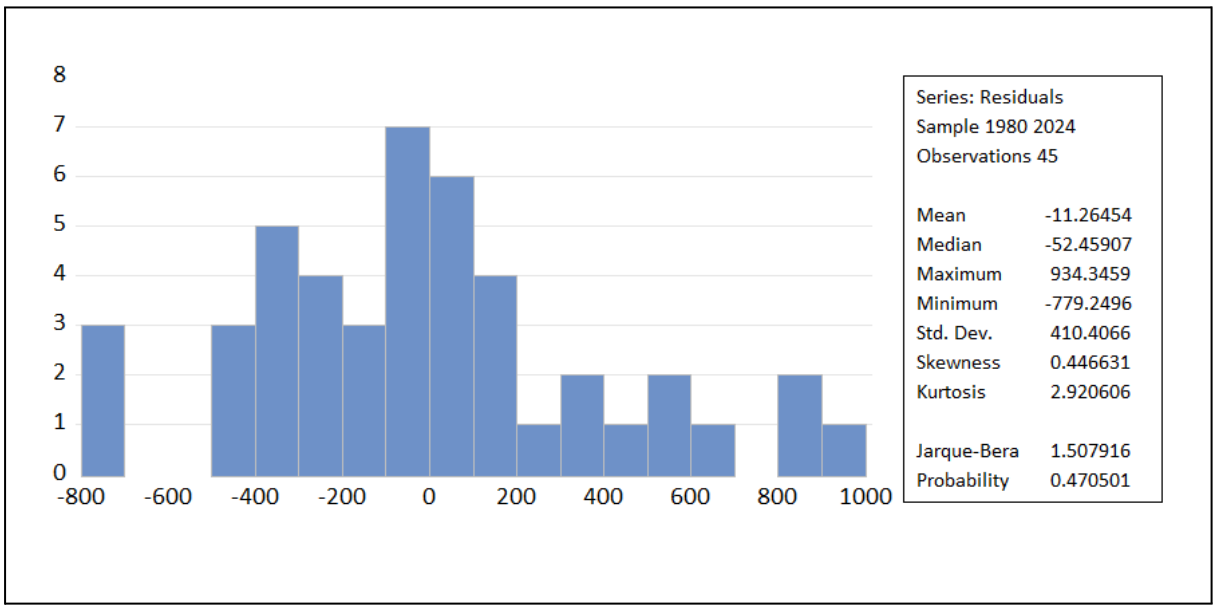
Inverted AR Roots	.55-.13i	.55+.13i

Date: 11/05/26 Time: 12:21
Sample: 1980 2024
Q-statistic probabilities adjusted for 2 ARMA terms

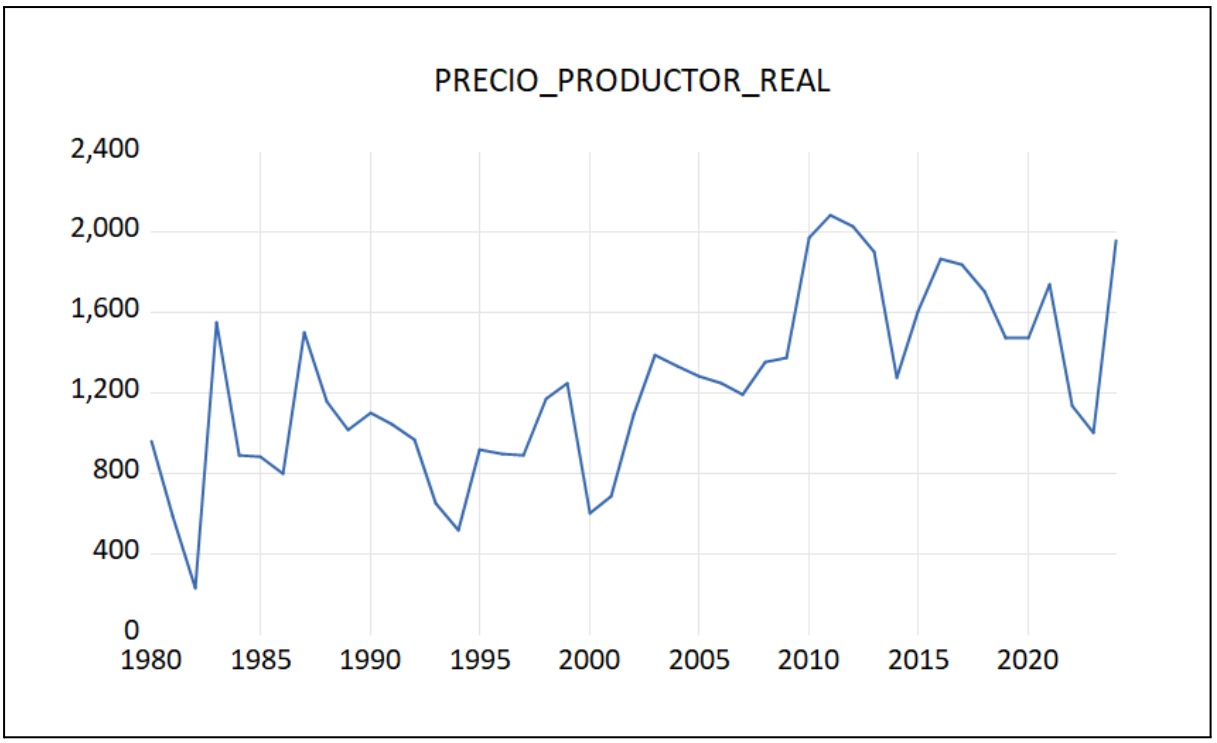


	AC	PAC	Q-Stat	Prob
1	-0.014	-0.014	0.0097	
2	-0.208	-0.209	2.1430	
3	-0.076	-0.086	2.4339	0.119
4	0.149	0.107	3.5762	0.167
5	0.128	0.108	4.4365	0.218
6	-0.134	-0.090	5.4146	0.247
7	0.136	0.209	6.4502	0.265
8	-0.025	-0.066	6.4863	0.371
9	-0.115	-0.112	7.2625	0.402
10	-0.119	-0.114	8.1226	0.422
11	-0.018	-0.097	8.1431	0.520
12	0.015	-0.107	8.1571	0.613
13	0.096	0.164	8.7663	0.643
14	0.034	0.054	8.8429	0.716
15	-0.238	-0.191	12.835	0.461
16	-0.159	-0.122	14.670	0.401
17	0.019	-0.082	14.698	0.473
18	0.182	0.034	17.296	0.367
19	-0.141	-0.134	18.912	0.334
20	-0.185	-0.140	21.802	0.241





De este modo incorporando la variable de intervención para el año 2024, el modelo que reporta mejor criterio de información, con coeficientes significativos y residuos ruido blanco, es el modelo **AR(2)**.


PRECIO_PRODUCTOR_REAL
2,400
2,000
1,600
1,200
800
400
0
1980
1985
1990
1995
2000
2005
2010
2015
2020

Null Hypothesis: PRECIO PRODUCTOR REAL has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-4.466497	0.0047
Test critical values:	1% level	-4.180911	
	5% level	-3.515523	
	10% level	-3.188259	

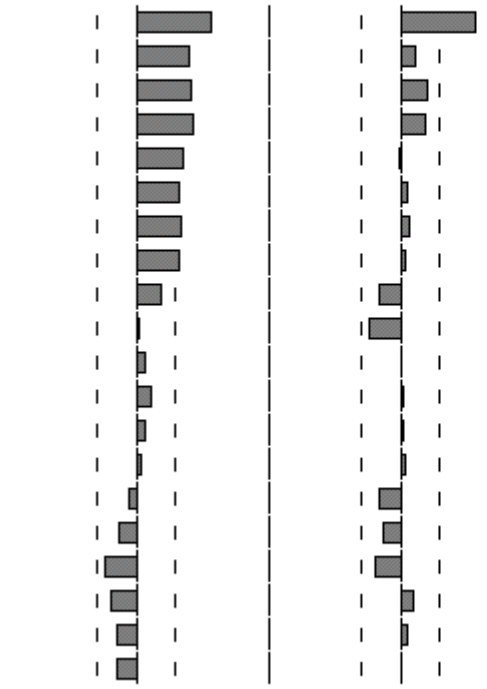
*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO PRODUCTOR REAL)
Method: Least Squares
Date: 11/05/26 Time: 12:25
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO PRODUCTOR REAL(-1)	-0.656128	0.146900	-4.466497	0.0001
C	477.4899	148.3321	3.219060	0.0025
@TREND("1980")	15.36929	4.988005	3.081249	0.0037

R-squared	0.328809	Mean dependent var	22.69169
Adjusted R-squared	0.296068	S.D. dependent var	390.7237
S.E. of regression	327.8197	Akaike info criterion	14.48855
Sum squared resid	4406097.	Schwarz criterion	14.61020
Log likelihood	-315.7481	Hannan-Quinn criter.	14.53366
F-statistic	10.04272	Durbin-Watson stat	1.850384
Prob(F-statistic)	0.000282		

Date: 11/05/26 Time: 12:25
Sample: 1980 2024
Included observations: 45

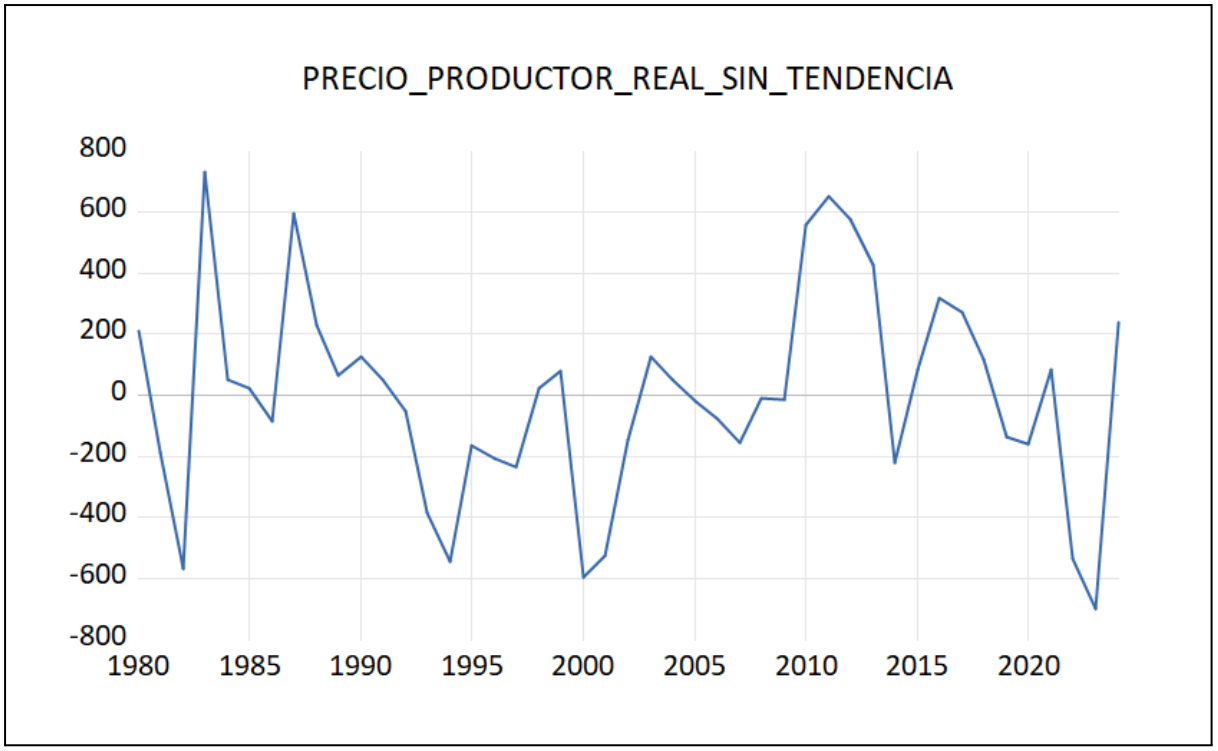


Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.588	0.588	16.606	0.000
		2	0.422	0.117	25.374	0.000
		3	0.429	0.218	34.641	0.000
		4	0.458	0.190	45.468	0.000
		5	0.368	-0.007	52.644	0.000
		6	0.335	0.060	58.738	0.000
		7	0.349	0.073	65.530	0.000
		8	0.337	0.033	72.019	0.000
		9	0.190	-0.166	74.139	0.000
		10	0.031	-0.248	74.197	0.000
		11	0.076	0.012	74.552	0.000
		12	0.114	0.030	75.391	0.000
		13	0.071	0.021	75.729	0.000
		14	0.039	0.041	75.831	0.000
		15	-0.054	-0.168	76.035	0.000
		16	-0.131	-0.132	77.284	0.000
		17	-0.244	-0.190	81.770	0.000
		18	-0.189	0.093	84.565	0.000
		19	-0.148	0.051	86.349	0.000
		20	-0.141	0.012	88.025	0.000

Dependent Variable: PRECIO PRODUCTOR REAL
Method: Least Squares
Date: 11/05/26 Time: 12:26
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	752.6148	100.4303	7.493902	0.0000
@TREND	22.00324	3.931145	5.597159	0.0000

R-squared	0.421485	Mean dependent var	1236.686
Adjusted R-squared	0.408031	S.D. dependent var	445.1336
S.E. of regression	342.4837	Akaike info criterion	14.55375
Sum squared resid	5043689.	Schwarz criterion	14.63405
Log likelihood	-325.4594	Hannan-Quinn criter.	14.58369
F-statistic	31.32819	Durbin-Watson stat	1.301550
Prob(F-statistic)	0.000001		



Date: 11/05/26 Time: 12:27
Sample: 1980 2024
Included observations: 45

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.339	0.339	5.5253	0.019
		2	0.050	-0.073	5.6493	0.059
		3	0.111	0.134	6.2699	0.099
		4	0.142	0.071	7.3151	0.120
		5	0.009	-0.070	7.3192	0.198
		6	-0.023	-0.004	7.3473	0.290
		7	0.062	0.060	7.5627	0.373
		8	0.049	-0.002	7.6999	0.463
		9	-0.158	-0.186	9.1598	0.423
		10	-0.393	-0.337	18.476	0.047
		11	-0.314	-0.154	24.604	0.010
		12	-0.175	-0.059	26.565	0.009
		13	-0.152	-0.015	28.086	0.009
		14	-0.068	0.102	28.398	0.013
		15	-0.078	-0.065	28.821	0.017
		16	-0.153	-0.128	30.526	0.015
		17	-0.286	-0.240	36.716	0.004
		18	-0.142	0.010	38.297	0.004
		19	-0.041	-0.049	38.435	0.005
		20	-0.001	-0.084	38.435	0.008

Dependent Variable: PRECIO PRODUCTOR REAL SIN TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 12:28
Sample: 1980 2024
Included observations: 45
Convergence achieved after 11 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.338436	0.125622	2.694081	0.0100
SIGMASQ	98937.02	19980.65	4.951641	0.0000

R-squared	0.117280	Mean dependent var	5.94E-14
Adjusted R-squared	0.096751	S.D. dependent var	338.5695
S.E. of regression	321.7744	Akaike info criterion	14.43171
Sum squared resid	4452166.	Schwarz criterion	14.51200
Log likelihood	-322.7134	Hannan-Quinn criter.	14.46164
Durbin-Watson stat	1.870049		

Inverted AR Roots	.34

Date: 11/05/26 Time: 12:28
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.035	0.035	0.0593	
		2	-0.117	-0.119	0.7359	0.391
		3	0.053	0.063	0.8771	0.645
		4	0.137	0.120	1.8425	0.606
		5	-0.037	-0.035	1.9147	0.751
		6	-0.059	-0.032	2.1046	0.834
		7	0.067	0.051	2.3542	0.884
		8	0.101	0.077	2.9423	0.890
		9	-0.062	-0.046	3.1664	0.923
		10	-0.312	-0.300	9.0540	0.432
		11	-0.185	-0.229	11.191	0.343
		12	-0.038	-0.126	11.282	0.420
		13	-0.091	-0.096	11.826	0.460
		14	0.012	0.102	11.837	0.541
		15	-0.003	0.017	11.837	0.619
		16	-0.053	-0.049	12.042	0.676
		17	-0.247	-0.257	16.644	0.409
		18	-0.056	-0.063	16.891	0.462
		19	0.002	-0.044	16.892	0.531
		20	-0.004	-0.071	16.893	0.597

Dependent Variable: PRECIO PRODUCTOR REAL SIN TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 12:29
Sample: 1980 2024
Included observations: 45
Convergence achieved after 18 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.186682	0.138473	1.348149	0.1848
MA(10)	-0.758067	0.253559	-2.989705	0.0047
SIGMASQ	65292.94	18778.81	3.476948	0.0012

R-squared	0.417454	Mean dependent var	5.94E-14
Adjusted R-squared	0.389713	S.D. dependent var	338.5695
S.E. of regression	264.4933	Akaike info criterion	14.23734
Sum squared resid	2938182.	Schwarz criterion	14.35779
Log likelihood	-317.3402	Hannan-Quinn criter.	14.28224
Durbin-Watson stat	1.974574		

Inverted AR Roots	.19			
Inverted MA Roots	.97	.79+.57i	.79-.57i	.30-.93i
	.30+.93i	-.30-.93i	-.30+.93i	-.79-.57i
	-.79+.57i	-.97		

Dependent Variable: PRECIO PRODUCTOR REAL SIN TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 12:31
Sample: 1980 2024
Included observations: 45
Convergence achieved after 7 iterations
Coefficient covariance computed using outer product of gradients

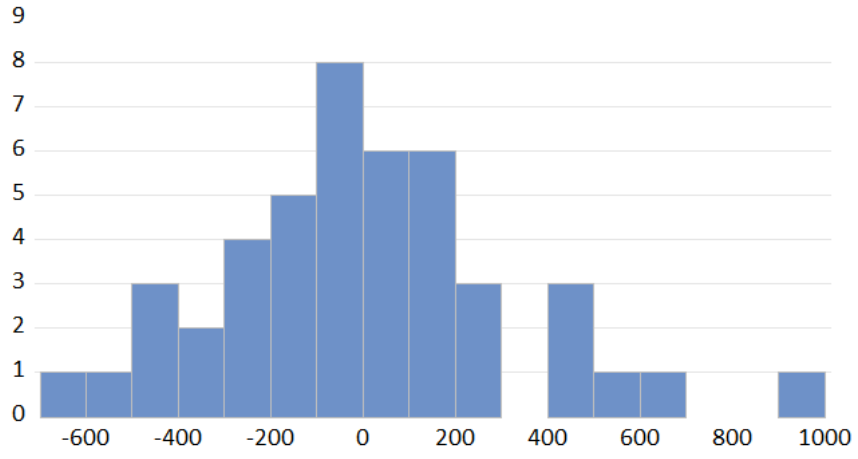
Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.377953	0.140547	2.689157	0.0102
SIGMASQ	97515.50	19075.29	5.112138	0.0000

R-squared	0.129963	Mean dependent var	5.94E-14
Adjusted R-squared	0.109729	S.D. dependent var	338.5695
S.E. of regression	319.4544	Akaike info criterion	14.41796
Sum squared resid	4388197.	Schwarz criterion	14.49825
Log likelihood	-322.4041	Hannan-Quinn criter.	14.44789
Durbin-Watson stat	1.923011		

Inverted MA Roots	-.38

Date: 11/05/26 Time: 12:31
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.014	0.014	0.0095	
		2	0.013	0.012	0.0172	0.896
		3	0.041	0.040	0.0999	0.951
		4	0.143	0.142	1.1499	0.765
		5	-0.033	-0.038	1.2068	0.877
		6	-0.034	-0.039	1.2704	0.938
		7	0.055	0.047	1.4410	0.963
		8	0.062	0.045	1.6594	0.976
		9	-0.069	-0.061	1.9369	0.983
		10	-0.302	-0.307	7.4378	0.592
		11	-0.204	-0.250	10.031	0.438
		12	-0.056	-0.079	10.232	0.510
		13	-0.117	-0.080	11.140	0.517
		14	0.008	0.114	11.145	0.599
		15	-0.043	0.008	11.275	0.664
		16	-0.056	-0.068	11.505	0.716
		17	-0.249	-0.257	16.207	0.439
		18	-0.056	-0.080	16.454	0.492
		19	-0.025	-0.021	16.504	0.557
		20	-0.002	-0.069	16.505	0.623

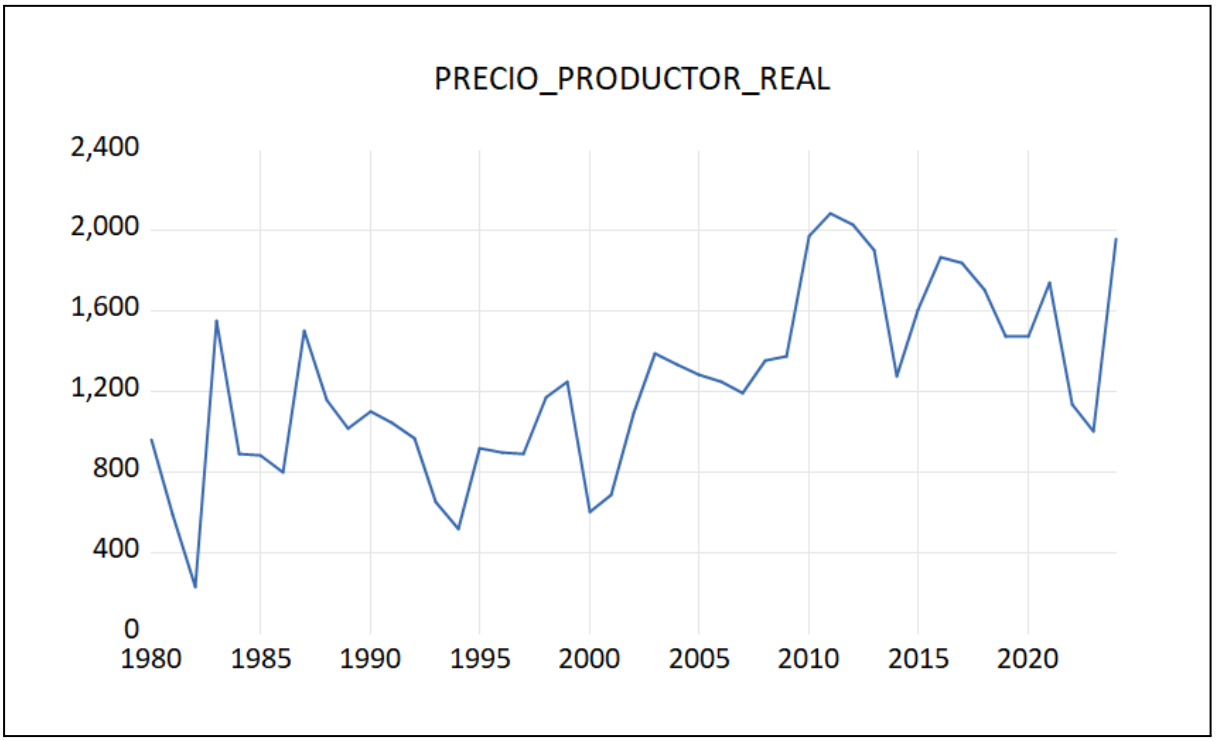


Series: Residuals
Sample 1980 2024
Observations 45

Mean	2.457488
Median	-10.63179
Maximum	911.0696
Minimum	-612.7970
Std. Dev.	315.7936
Skewness	0.431264
Kurtosis	3.527354
Jarque-Bera	1.916355
Probability	0.383591

Luego de realizar las estimaciones se ve como resulta significativo el rezago número diez en el modelo AR y MA, sin embargo incluirlo en la estimación genera problemas de significatividad en los coeficientes y de raíces cerca de la unidad, además si se diferencia la serie una vez, las estimaciones de los modelos arrojan una mayor varianza que los modelo en nivel, por lo tanto decido eliminar el componente MA(10) de ambas estimaciones y comparar los modelos AR(1) y MA(1) según el AIC, en donde gana el MA(1) por tener un menor criterio (14.41<14.43). Además se puede ver como el modelo MA(1) tiene raíces dentro del círculo unitario, y los residuos se comportan normalmente.

Incorporando una variable de intervencion:



Al igual que ocurrió con el precio internacional, el fuerte aumento del precio del cacao provocó que el gobierno Ghanes deba ajustar (aumentar) durante el 2024 el precio que establecido antes de la cosecha, además lo mismo ocurrió en el año 1983, con un fuerte aumento como incentivo a los productores de cacao como consecuencia de las reformas económicas (ERP).
Por lo tanto incorporo en la estimación ambas variables de intervención

Dependent Variable: PRECIO PRODUCTOR REAL
Method: Least Squares
Date: 11/05/26 Time: 15:35
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
VI 1983	780.4922	340.4163	2.292758	0.0271
VI 2024	235.9633	343.4208	0.687097	0.4959
C	702.0902	100.8977	6.958439	0.0000
@TREND	23.27309	3.995232	5.825216	0.0000

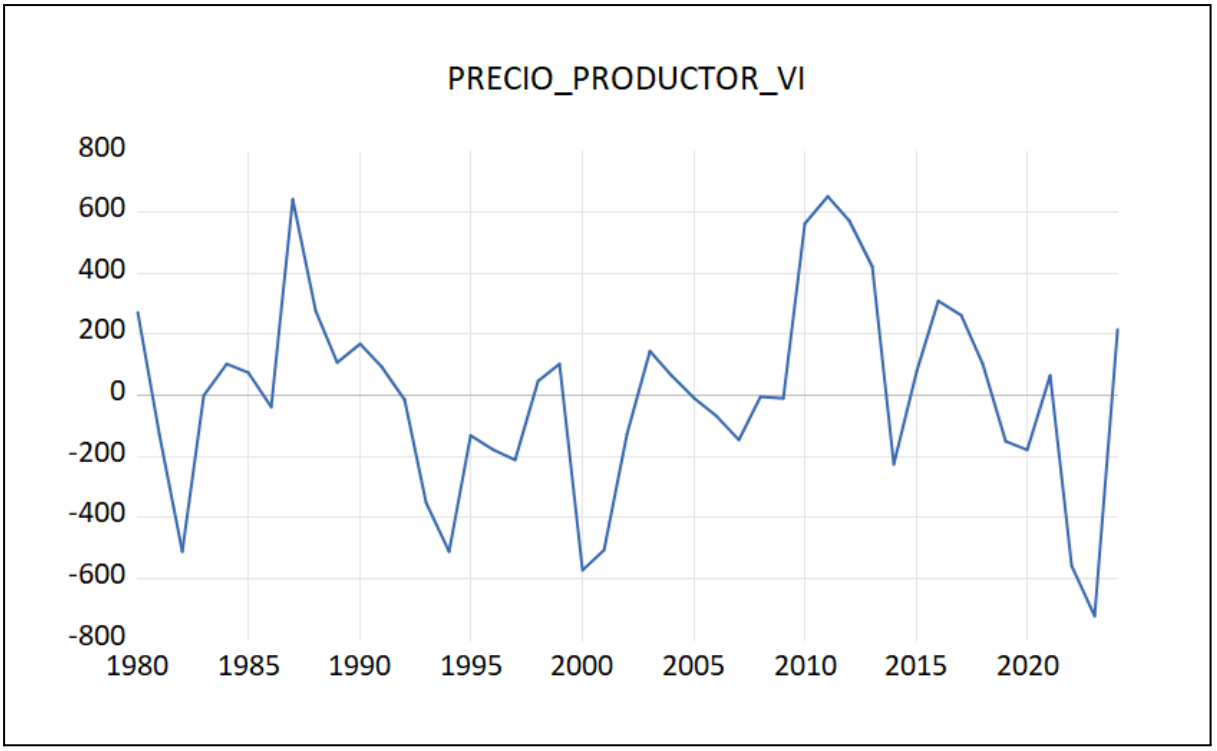
R-squared	0.493706	Mean dependent var	1236.686
Adjusted R-squared	0.456660	S.D. dependent var	445.1336
S.E. of regression	328.1151	Akaike info criterion	14.50929
Sum squared resid	4414040.	Schwarz criterion	14.66989
Log likelihood	-322.4591	Hannan-Quinn criter.	14.56916
F-statistic	13.32686	Durbin-Watson stat	0.974221
Prob(F-statistic)	0.000003		

Como sólo resulta significativa la variable de intervención de 1983, es la que dejo en el modelo.

Dependent Variable: PRECIO PRODUCTOR REAL
Method: Least Squares
Date: 11/05/26 Time: 15:36
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
VI 1983	788.8252	338.0554	2.333420	0.0245
C	691.6428	99.11655	6.978076	0.0000
@TREND	23.97790	3.836958	6.249195	0.0000

R-squared	0.487876	Mean dependent var	1236.686
Adjusted R-squared	0.463489	S.D. dependent var	445.1336
S.E. of regression	326.0465	Akaike info criterion	14.47630
Sum squared resid	4464867.	Schwarz criterion	14.59674
Log likelihood	-322.7167	Hannan-Quinn criter.	14.52120
F-statistic	20.00569	Durbin-Watson stat	1.048355
Prob(F-statistic)	0.000001		


PRECIO_PRODUCTOR_VI
800
600
400
200
0
-200
-400
-600
-800
1980
1985
1990
1995
2000
2005
2010
2015
2020

Null Hypothesis: PRECIO PRODUCTOR VI has a unit root
Exogenous: None
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.967477	0.0002
Test critical values:	1% level	-2.618579	
	5% level	-1.948495	
	10% level	-1.612135	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO PRODUCTOR VI)
Method: Least Squares
Date: 11/05/26 Time: 15:38
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO PRODUCTOR VI(-1)	-0.532803	0.134293	-3.967477	0.0003

R-squared	0.267960	Mean dependent var	-1.286211
Adjusted R-squared	0.267960	S.D. dependent var	329.9292
S.E. of regression	282.2853	Akaike info criterion	14.14618
Sum squared resid	3426454.	Schwarz criterion	14.18673
Log likelihood	-310.2159	Hannan-Quinn criter.	14.16122
Durbin-Watson stat	1.655973		

Date: 11/05/26 Time: 15:38
Sample: 1980 2024
Included observations: 45

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.462	0.462	10.275	0.001
		2	0.072	-0.180	10.531	0.005
		3	0.093	0.180	10.968	0.012
		4	0.046	-0.096	11.079	0.026
		5	-0.042	-0.019	11.174	0.048
		6	-0.051	-0.025	11.315	0.079
		7	0.035	0.085	11.383	0.123
		8	0.026	-0.042	11.423	0.179
		9	-0.191	-0.236	13.568	0.139
		10	-0.398	-0.282	23.163	0.010
		11	-0.286	-0.002	28.242	0.003
		12	-0.195	-0.112	30.672	0.002
		13	-0.157	0.028	32.308	0.002
		14	-0.048	0.009	32.465	0.003
		15	-0.094	-0.196	33.083	0.005
		16	-0.186	-0.143	35.613	0.003
		17	-0.214	-0.127	39.085	0.002
		18	-0.060	0.122	39.369	0.003
		19	-0.014	-0.170	39.385	0.004
		20	-0.017	-0.108	39.410	0.006

Dependent Variable: PRECIO PRODUCTOR VI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 15:40
Sample: 1980 2024
Included observations: 45
Convergence achieved after 35 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.066811	0.236201	0.282856	0.7787
MA(1)	0.583150	0.225476	2.586313	0.0133
SIGMASQ	70102.35	13363.45	5.245828	0.0000

R-squared	0.293460	Mean dependent var	9.35E-14
Adjusted R-squared	0.259815	S.D. dependent var	318.5502
S.E. of regression	274.0614	Akaike info criterion	14.13996
Sum squared resid	3154606.	Schwarz criterion	14.26040
Log likelihood	-315.1490	Hannan-Quinn criter.	14.18486
Durbin-Watson stat	1.909410		

Inverted AR Roots	.07
Inverted MA Roots	-.58

Dependent Variable: PRECIO PRODUCTOR VI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 15:40
Sample: 1980 2024
Included observations: 45
Convergence achieved after 7 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.464593	0.120781	3.846577	0.0004
SIGMASQ	77433.23	15884.10	4.874890	0.0000

R-squared	0.219575	Mean dependent var	9.35E-14
Adjusted R-squared	0.201425	S.D. dependent var	318.5502
S.E. of regression	284.6661	Akaike info criterion	14.18934
Sum squared resid	3484495.	Schwarz criterion	14.26964
Log likelihood	-317.2602	Hannan-Quinn criter.	14.21927
Durbin-Watson stat	1.695829		

Inverted AR Roots	.46

Date: 11/05/26 Time: 15:41
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.100	0.100	0.4854	
		2	-0.228	-0.240	3.0344	0.082
		3	0.055	0.115	3.1873	0.203
		4	0.045	-0.036	3.2921	0.349
		5	-0.067	-0.031	3.5287	0.474
		6	-0.081	-0.075	3.8809	0.567
		7	0.063	0.063	4.0998	0.663
		8	0.137	0.101	5.1754	0.639
		9	-0.068	-0.069	5.4471	0.709
		10	-0.333	-0.299	12.132	0.206
		11	-0.104	-0.089	12.809	0.235
		12	-0.035	-0.165	12.886	0.301
		13	-0.090	-0.053	13.416	0.340
		14	0.089	0.097	13.957	0.377
		15	0.009	-0.096	13.963	0.452
		16	-0.106	-0.125	14.778	0.468
		17	-0.184	-0.239	17.336	0.364
		18	0.038	0.088	17.447	0.424
		19	0.022	-0.088	17.488	0.490
		20	-0.053	-0.097	17.725	0.541

Dependent Variable: PRECIO PRODUCTOR VI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 15:42
Sample: 1980 2024
Included observations: 45
Convergence achieved after 12 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.358847	0.115699	3.101552	0.0034
AR(10)	-0.419358	0.127957	-3.277327	0.0021
SIGMASQ	61265.33	16870.37	3.631535	0.0008

R-squared	0.382526	Mean dependent var	9.35E-14
Adjusted R-squared	0.353122	S.D. dependent var	318.5502
S.E. of regression	256.2058	Akaike info criterion	14.05054
Sum squared resid	2756940.	Schwarz criterion	14.17099
Log likelihood	-313.1372	Hannan-Quinn criter.	14.09544
Durbin-Watson stat	1.722941		

Inverted AR Roots	.92-.28i	.92+.28i	.58+.74i	.58-.74i
	.03+.91i	.03-.91i	-.51-.74i	-.51+.74i
	-.84-.28i	-.84+.28i		

Date: 11/05/26 Time: 15:43
Sample: 1980 2024
Q-statistic probabilities adjusted for 2 ARMA terms

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.105	0.105	0.5274	
		2	-0.224	-0.237	2.9923	
		3	0.079	0.142	3.3031	0.069
		4	0.077	-0.009	3.6097	0.164
		5	-0.129	-0.096	4.4857	0.214
		6	-0.167	-0.137	5.9918	0.200
		7	0.007	-0.011	5.9943	0.307
		8	0.130	0.095	6.9633	0.324
		9	-0.072	-0.078	7.2671	0.402
		10	-0.121	-0.061	8.1545	0.419
		11	0.043	-0.017	8.2687	0.507
		12	0.046	-0.010	8.4020	0.590
		13	-0.049	-0.006	8.5619	0.662
		14	0.070	0.110	8.8963	0.712
		15	0.010	-0.076	8.9028	0.780
		16	-0.168	-0.174	10.950	0.690
		17	-0.261	-0.269	16.092	0.376
		18	0.064	0.096	16.411	0.425
		19	0.021	-0.079	16.448	0.492
		20	-0.176	-0.107	19.077	0.387

Dependent Variable: PRECIO PRODUCTOR VI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 15:44
Sample: 1980 2024
Included observations: 45
Convergence achieved after 29 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.622026	0.153787	4.044721	0.0002
SIGMASQ	70237.87	13432.52	5.228941	0.0000

R-squared	0.292094	Mean dependent var	9.35E-14
Adjusted R-squared	0.275631	S.D. dependent var	318.5502
S.E. of regression	271.1176	Akaike info criterion	14.09728
Sum squared resid	3160704.	Schwarz criterion	14.17758
Log likelihood	-315.1888	Hannan-Quinn criter.	14.12721
Durbin-Watson stat	1.865293		

Inverted MA Roots	-.62

Date: 11/05/26 Time: 15:44
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.032	0.032	0.0479	
		2	0.008	0.007	0.0514	0.821
		3	0.027	0.026	0.0873	0.957
		4	0.067	0.066	0.3212	0.956
		5	-0.069	-0.074	0.5742	0.966
		6	-0.040	-0.037	0.6607	0.985
		7	0.030	0.030	0.7101	0.994
		8	0.055	0.053	0.8803	0.997
		9	-0.063	-0.056	1.1128	0.997
		10	-0.307	-0.312	6.7987	0.658
		11	-0.148	-0.162	8.1688	0.612
		12	-0.042	-0.034	8.2794	0.688
		13	-0.150	-0.124	9.7574	0.637
		14	0.095	0.152	10.375	0.663
		15	-0.073	-0.120	10.750	0.706
		16	-0.067	-0.129	11.081	0.747
		17	-0.198	-0.224	14.040	0.596
		18	0.031	0.029	14.115	0.659
		19	-0.022	-0.002	14.153	0.719
		20	-0.031	-0.147	14.236	0.770

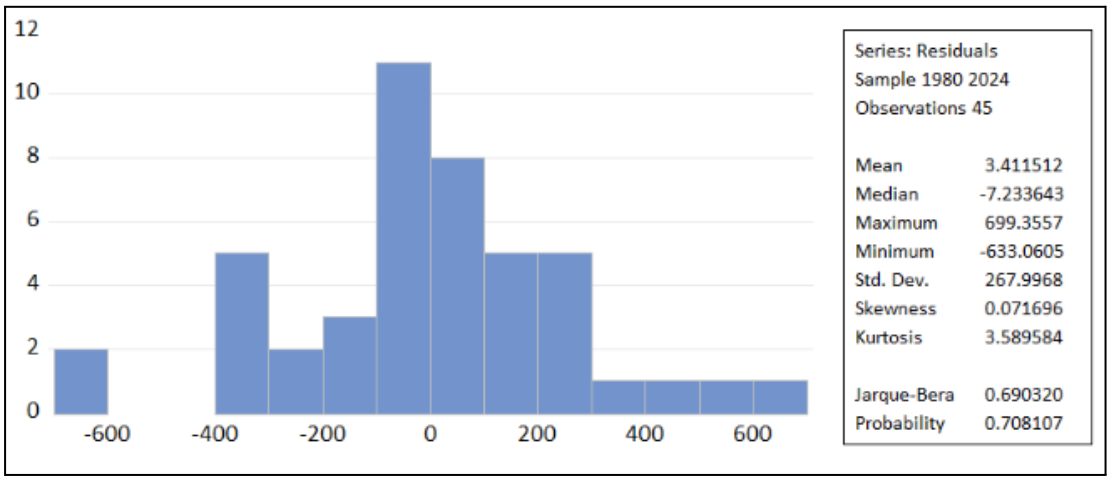
Dependent Variable: PRECIO PRODUCTOR VI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 15:44
Sample: 1980 2024
Included observations: 45
Convergence achieved after 28 iterations
Coefficient covariance computed using outer product of gradients

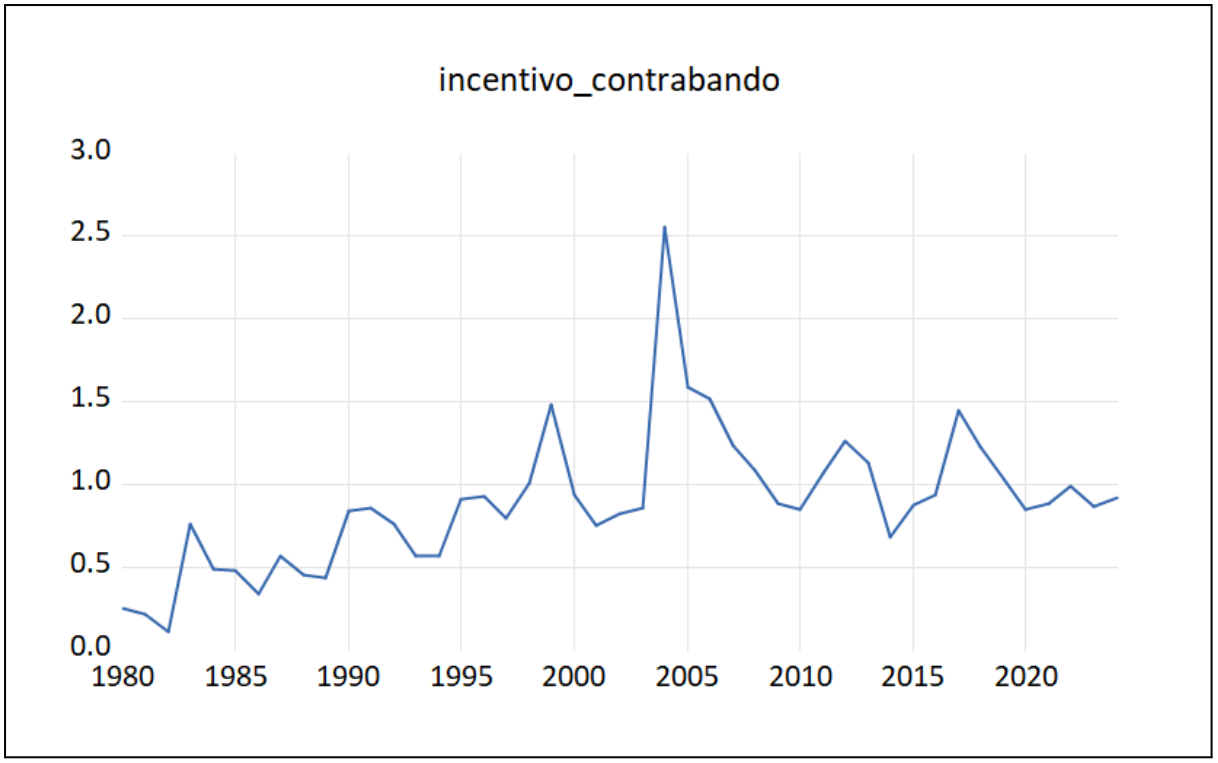
Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(10)	-0.464290	0.163360	-2.842135	0.0069
MA(1)	0.540656	0.158322	3.414922	0.0014
SIGMASQ	56960.11	16349.65	3.483873	0.0012

R-squared	0.425917	Mean dependent var	9.35E-14
Adjusted R-squared	0.398580	S.D. dependent var	318.5502
S.E. of regression	247.0398	Akaike info criterion	13.98300
Sum squared resid	2563205.	Schwarz criterion	14.10344
Log likelihood	-311.6175	Hannan-Quinn criter.	14.02790
Durbin-Watson stat	1.916948		

Inverted AR Roots	.88-.29i	.88+.29i	.54-.75i	.54+.75i
	-.00+.93i	-.00-.93i	-.54+.75i	-.54-.75i
	-.88+.29i	-.88-.29i		
Inverted MA Roots	-.54			

Nuevamente, al igual que antes de incorporar la variable de intervención aparece el rezago 10 casi significativo, si se lo incorpora, por un lado mejora el AIC, pero las raíces quedan muy cerca de la unidad, también rompe con la parsimonia de un modelo que de por si ya posee pocas observaciones (45), por lo que incluirlo limitaría mucho los grados de libertad, además de que económicamente hablando implicaría suponer que el precio del productor está determinado por lo ocurrido hace 10 años, lo cual no le encuentro mucho sentido, por lo tanto decido no incluirlo en el modelo, a pesar de que los residuos no queden completamente ruido blanco; además diferenciar la serie no es una opción debido a que aumenta la varianza en las estimaciones y el ADF fue rechazado anteriormente.
Por lo tanto nuevamente comparo los modelos AR(1) y MA(1), ambos sin el componente del rezago 10, y el que tiene menor AIC es el modelo **MA(1),** que posee **residuos aproximadamente ruido blanco**, debido a la presencia de la autocorrelación en el décimo rezago, lo cual debe ser tenido en cuenta para la aplicación de próximos modelos como por ejemplo el Modelo de Función de Transferencia**.**





Se nota una leve tendencia entre 1980 y principios de los 2000 que termina con la presencia de un salto abrupto en el año 2004 que puede deberse a la guerra civil ocurrida en Costa de Marfil, que ocurrió entre los años 2002-2007, pero que tuvo su apogeo en el 2004. A partir del 2010 se observa cierta estabilidad.

Analizo los test de Dickey Fuller para verificar la existencia de raíz unitaria, empezando con el modelo con tendencia determinística e intercepto:

Null Hypothesis: INCENTIVO CONTRABANDO has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.941365	0.0184
Test critical values:	1% level	-4.180911	
	5% level	-3.515523	
	10% level	-3.188259	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(INCENTIVO CONTRABANDO)
Method: Least Squares
Date: 10/14/26 Time: 19:10
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
INCENTIVO CONTRABANDO(-1)	-0.553339	0.140393	-3.941365	0.0003
C	0.332058	0.123311	2.692847	0.0102
@TREND("1980")	0.007811	0.004657	1.677212	0.1011

R-squared	0.277855	Mean dependent var	0.015094
Adjusted R-squared	0.242628	S.D. dependent var	0.381944
S.E. of regression	0.332394	Akaike info criterion	0.700757
Sum squared resid	4.529927	Schwarz criterion	0.822406
Log likelihood	-12.41665	Hannan-Quinn criter.	0.745870
F-statistic	7.887653	Durbin-Watson stat	2.020398
Prob(F-statistic)	0.001264		

Como la tendencia determinística no resulta significativa (p-valor = 0.1011), vuelvo a hacer el test con intercepto.

Null Hypothesis: INCENTIVO CONTRABANDO has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.525035	0.0118
Test critical values:	1% level	-3.588509	
	5% level	-2.929734	
	10% level	-2.603064	

*MacKinnon (1996) one-sided p-values.

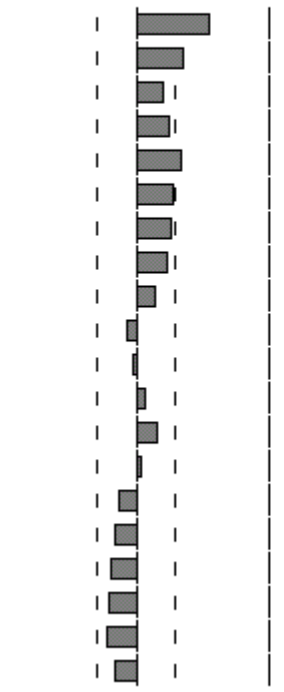
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(INCENTIVO CONTRABANDO)
Method: Least Squares
Date: 10/14/26 Time: 19:10
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
INCENTIVO CONTRABANDO(-1)	-0.428289	0.121499	-3.525035	0.0010
C	0.396459	0.119683	3.312581	0.0019

R-squared	0.228308	Mean dependent var	0.015094
Adjusted R-squared	0.209935	S.D. dependent var	0.381944
S.E. of regression	0.339493	Akaike info criterion	0.721662
Sum squared resid	4.840729	Schwarz criterion	0.802761
Log likelihood	-13.87656	Hannan-Quinn criter.	0.751737
F-statistic	12.42587	Durbin-Watson stat	2.150524
Prob(F-statistic)	0.001038		

El intercepto resulta significativo por lo que analizo el p-valor asociado al estadístico tau, como 0.0118<0.05 rechazo la hipótesis nula de raíz unitaria, por lo que proceso a analizar el correlograma de la serie en niveles:

Date: 10/14/26 Time: 19:11
Sample: 1980 2024
Included observations: 45



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.572	0.572	15.706	0.000
		2	0.373	0.069	22.546	0.000
		3	0.211	-0.039	24.795	0.000
		4	0.266	0.212	28.439	0.000
		5	0.350	0.198	34.927	0.000
		6	0.289	-0.049	39.459	0.000
		7	0.274	0.095	43.653	0.000
		8	0.251	0.088	47.244	0.000
		9	0.144	-0.162	48.462	0.000
		10	-0.075	-0.331	48.806	0.000
		11	-0.025	0.169	48.846	0.000
		12	0.071	0.109	49.166	0.000
		13	0.156	-0.065	50.780	0.000
		14	0.048	-0.097	50.935	0.000
		15	-0.126	-0.061	52.054	0.000
		16	-0.166	-0.071	54.060	0.000
		17	-0.190	-0.131	56.798	0.000
		18	-0.218	-0.090	60.527	0.000
		19	-0.226	-0.021	64.693	0.000
		20	-0.171	-0.035	67.165	0.000

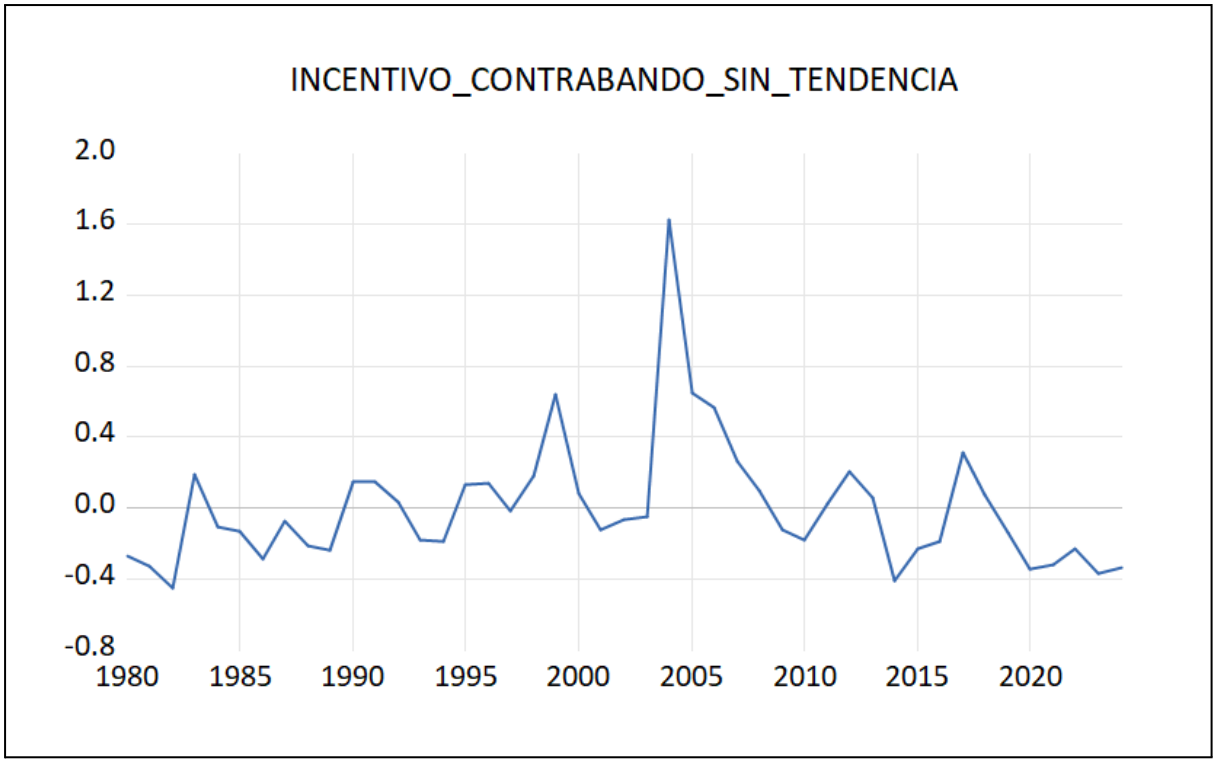
Dependent Variable: INCENTIVO CONTRABANDO
Method: Least Squares
Date: 10/14/26 Time: 19:12
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
@TREND	0.016547	0.004190	3.949091	0.0003
C	0.527008	0.107046	4.923183	0.0000

R-squared	0.266153	Mean dependent var	0.891044
Adjusted R-squared	0.249087	S.D. dependent var	0.421261
S.E. of regression	0.365045	Akaike info criterion	0.865833
Sum squared resid	5.730078	Schwarz criterion	0.946129
Log likelihood	-17.48123	Hannan-Quinn criter.	0.895766
F-statistic	15.59532	Durbin-Watson stat	1.094744
Prob(F-statistic)	0.000286		

Notar que en el ADF la tendencia no resultaba significativa, sin embargo al estimarla junto al intercepto tiene un p-valor = 0.0003.

Como tanto la tendencia como el intercepto resultan significativos, hago la serie de los residuos para evaluar la serie sin tendencia:



Date: 10/14/26 Time: 19:23
Sample: 1980 2024
Included observations: 45

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.436	0.436	9.1448	0.002
		2	0.186	-0.005	10.854	0.004
		3	0.005	-0.093	10.855	0.013
		4	0.070	0.126	11.105	0.025
		5	0.192	0.168	13.048	0.023
		6	0.130	-0.050	13.958	0.030
		7	0.155	0.110	15.289	0.032
		8	0.170	0.123	16.937	0.031
		9	0.049	-0.130	17.077	0.048
		10	-0.227	-0.338	20.199	0.027
		11	-0.174	0.093	22.082	0.024
		12	-0.030	0.094	22.140	0.036
		13	0.117	-0.015	23.048	0.041
		14	0.002	-0.128	23.048	0.059
		15	-0.215	-0.136	26.320	0.035
		16	-0.269	-0.114	31.598	0.011
		17	-0.288	-0.144	37.884	0.003
		18	-0.293	-0.122	44.591	0.000
		19	-0.263	-0.057	50.210	0.000
		20	-0.171	-0.085	52.686	0.000

Analizando el correlograma, tanto la FAC como la FACP resultan significativas solamente en el primer rezago (y el rezago 10 de la FACP)

Dependent Variable: INCENTIVO CONTRABANDO SIN TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 11:50
Sample: 1980 2024
Included observations: 45
Convergence achieved after 33 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.451557	0.442462	1.020556	0.3133
MA(1)	-0.013072	0.608852	-0.021469	0.9830
SIGMASQ	0.102293	0.008015	12.76265	0.0000

R-squared	0.196661	Mean dependent var	1.78E-16
Adjusted R-squared	0.158407	S.D. dependent var	0.360873
S.E. of regression	0.331059	Akaike info criterion	0.696107
Sum squared resid	4.603194	Schwarz criterion	0.816551
Log likelihood	-12.66240	Hannan-Quinn criter.	0.741007
Durbin-Watson stat	1.971198		

Inverted AR Roots	.45
Inverted MA Roots	.01

Dependent Variable: INCENTIVO CONTRABANDO SIN TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 14:50
Sample: 1980 2024
Included observations: 45
Convergence achieved after 7 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.440912	0.265595	1.660092	0.1042
SIGMASQ	0.102298	0.007957	12.85634	0.0000

R-squared	0.196622	Mean dependent var	1.78E-16
Adjusted R-squared	0.177938	S.D. dependent var	0.360873
S.E. of regression	0.327195	Akaike info criterion	0.651707
Sum squared resid	4.603421	Schwarz criterion	0.732003
Log likelihood	-12.66341	Hannan-Quinn criter.	0.681641
Durbin-Watson stat	1.976939		

Inverted AR Roots	.44

Date: 11/05/26 Time: 14:51
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.002	0.002	0.0002	
		2	0.043	0.043	0.0891	0.765
		3	-0.128	-0.128	0.9103	0.634
		4	0.001	-0.000	0.9103	0.823
		5	0.185	0.200	2.7234	0.605
		6	0.007	-0.012	2.7265	0.742
		7	0.069	0.051	2.9936	0.810
		8	0.128	0.189	3.9290	0.788
		9	0.107	0.107	4.6009	0.799
		10	-0.265	-0.337	8.8440	0.452
		11	-0.102	-0.083	9.4880	0.486
		12	-0.011	0.064	9.4960	0.576
		13	0.183	0.060	11.718	0.469
		14	0.053	-0.045	11.913	0.535
		15	-0.167	-0.086	13.877	0.459
		16	-0.116	-0.085	14.862	0.461
		17	-0.119	-0.134	15.939	0.457
		18	-0.130	-0.159	17.270	0.436
		19	-0.141	-0.090	18.884	0.399
		20	-0.127	-0.197	20.249	0.380

Dependent Variable: INCENTIVO CONTRABANDO SIN TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 14:52
Sample: 1980 2024
Included observations: 45
Convergence achieved after 9 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.438785	0.242707	1.807877	0.0778
AR(10)	-0.239242	0.339483	-0.704724	0.4849
SIGMASQ	0.093367	0.010379	8.995690	0.0000

R-squared	0.266762	Mean dependent var	1.78E-16
Adjusted R-squared	0.231846	S.D. dependent var	0.360873
S.E. of regression	0.316285	Akaike info criterion	0.621812
Sum squared resid	4.201511	Schwarz criterion	0.742256
Log likelihood	-10.99077	Hannan-Quinn criter.	0.666713
Durbin-Watson stat	1.971948		

Inverted AR Roots	.88+.26i	.88-.26i	.56-.69i	.56+.69i
	.04-.86i	.04+.86i	-.47+.70i	-.47-.70i
	-.79-.27i	-.79+.27i		

Dependent Variable: INCENTIVO CONTRABANDO SIN TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 14:51
Sample: 1980 2024
Included observations: 45
Convergence achieved after 6 iterations
Coefficient covariance computed using outer product of gradients

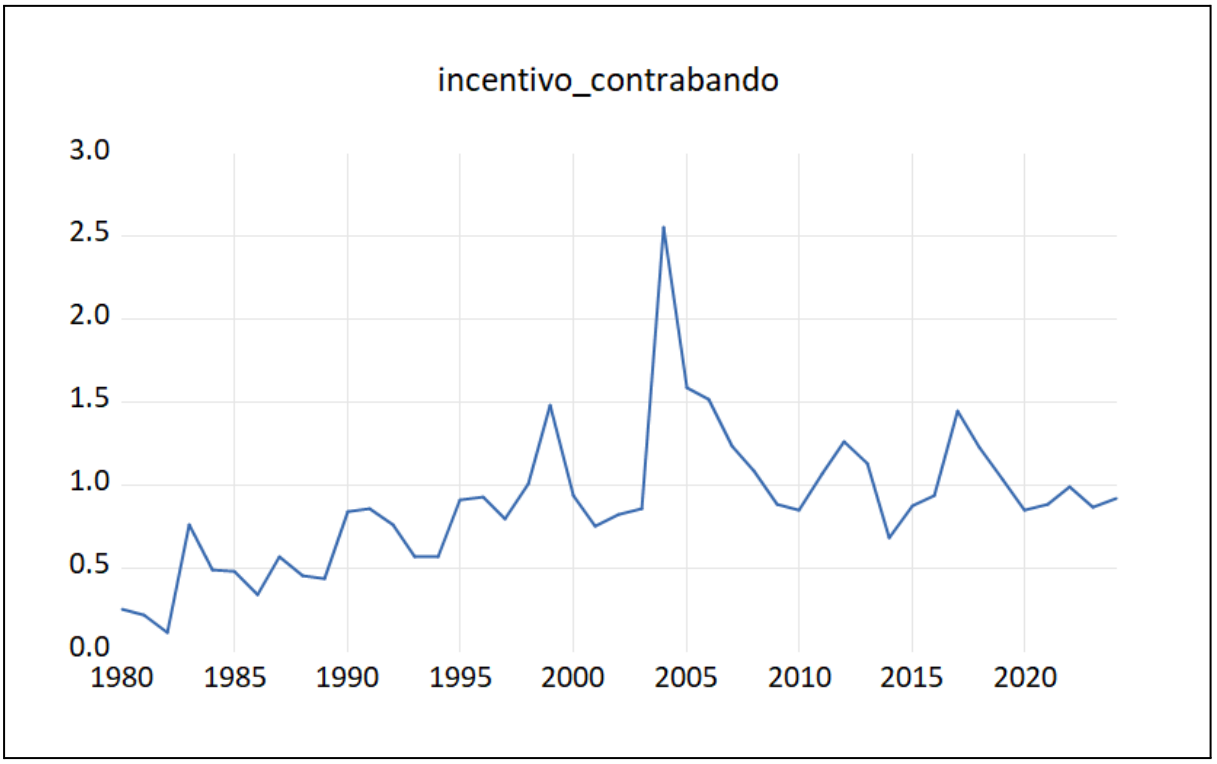
Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.359727	0.267820	1.343167	0.1863
SIGMASQ	0.107085	0.008679	12.33834	0.0000

R-squared	0.159029	Mean dependent var	1.78E-16
Adjusted R-squared	0.139471	S.D. dependent var	0.360873
S.E. of regression	0.334762	Akaike info criterion	0.695714
Sum squared resid	4.818830	Schwarz criterion	0.776010
Log likelihood	-13.65357	Hannan-Quinn criter.	0.725648
Durbin-Watson stat	1.809635		

Inverted MA Roots	-.36

La estimación del modelo resulta muy difícil, ya que ningún coeficiente resulta significativo, lo cual puede deberse a la presencia de el valor atípico en el año 2004.
Por lo tanto, procedo a incorporar una variable de intervención para solucionar este problema.

Incorporando una Variable de intervención:



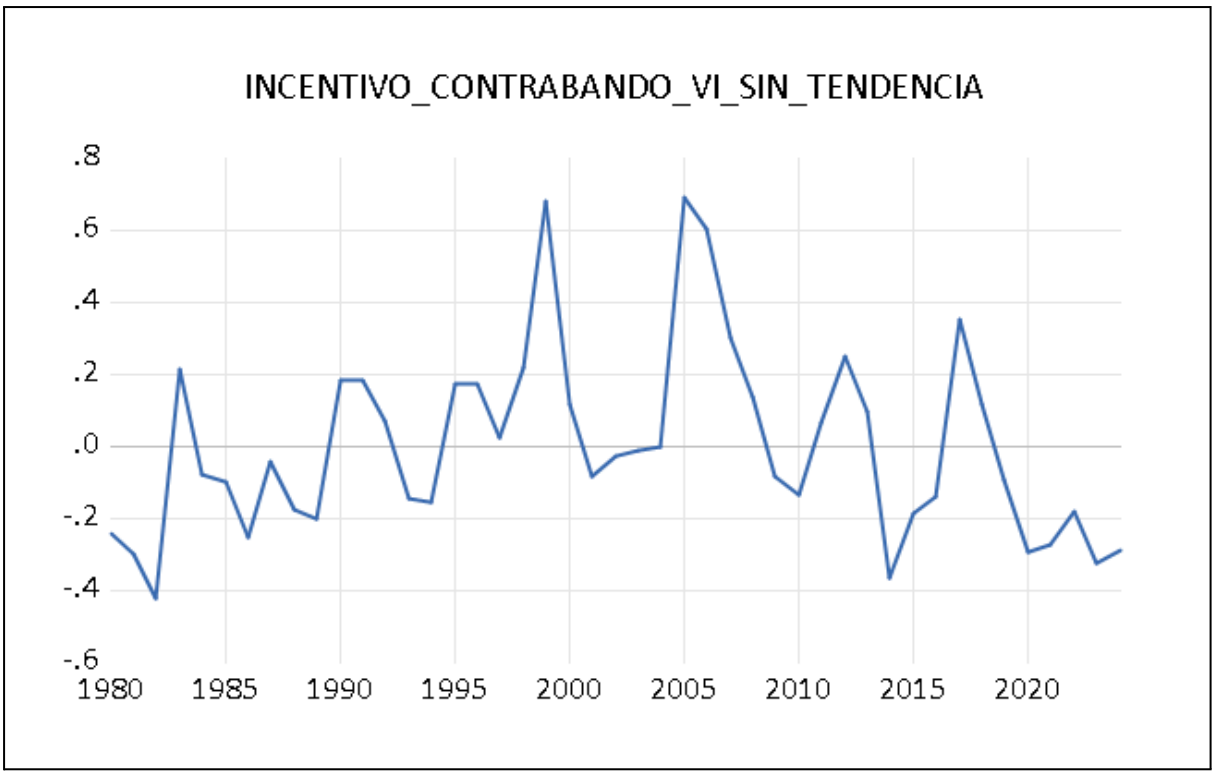
En este caso la variable de intervención se incorpora en el año 2004, debido a que fue el pico de la guerra civil de costa de marfil, que duró desde 2002 hasta 2007.
En ese año el precio de productor en costa de marfil fue extremadamente bajo, comparado con el de Ghana, lo que provoca que el incentivo de contrabando sea tan elevado.

Dependent Variable: INCENTIVO_CONTRABANDO
Method: Least Squares
Date: 11/02/26 Time: 13:16
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.499616	0.078658	6.351742	0.0000
@TREND	0.016108	0.003075	5.238586	0.0000
VI_2004	1.667653	0.270907	6.155822	0.0000

R-squared	0.614220	Mean dependent var	0.891044
Adjusted R-squared	0.595849	S.D. dependent var	0.421261
S.E. of regression	0.267808	Akaike info criterion	0.267244
Sum squared resid	3.012277	Schwarz criterion	0.387688
Log likelihood	-3.012993	Hannan-Quinn criter.	0.312144
F-statistic	33.43515	Durbin-Watson stat	0.984937
Prob(F-statistic)	0.000000		

La variable de impacto incorporada es un impulso aditivo (AO), el cual es una variable que toma valores 0 en todos los años, excepto en 2004, cuando vale 1, se estimó junto a la tendencia y el intercepto y los tres coeficientes resultaron significativos.



Null Hypothesis: INCENTIVO_CONTRABANDO_VI_SIN_TENDENCIA has a unit root
Exogenous: None
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.778555	0.0003
Test critical values:	1% level	-2.618579	
	5% level	-1.948495	
	10% level	-1.612135	

*MacKinnon (1996) one-sided p-values.

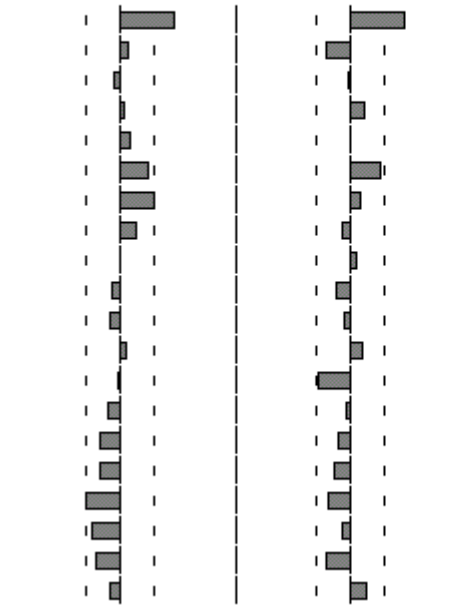
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(INCENTIVO_CONTRABANDO_VI_SIN_TENDENCIA
)
Method: Least Squares
Date: 11/02/26 Time: 13:18
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
INCENTIVO_CONTRABANDO_VI_SIN_TE	-0.502591	0.133011	-3.778555	0.0005

R-squared	0.249257	Mean dependent var	-0.001013
Adjusted R-squared	0.249257	S.D. dependent var	0.262672
S.E. of regression	0.227593	Akaike info criterion	-0.100047
Sum squared resid	2.227348	Schwarz criterion	-0.059497
Log likelihood	3.201025	Hannan-Quinn criter.	-0.085009
Durbin-Watson stat	1.814054		

El nuevo ADF sin el impacto de 2004 vuelve a dar una serie estacionaria en niveles.

Date: 11/02/26 Time: 13:18
Sample: 1980 2024
Included observations: 45



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.483	0.483	11.235	0.001
		2	0.074	-0.209	11.502	0.003
		3	-0.066	-0.010	11.719	0.008
		4	0.038	0.131	11.793	0.019
		5	0.088	-0.003	12.203	0.032
		6	0.256	0.274	15.758	0.015
		7	0.302	0.097	20.824	0.004
		8	0.141	-0.075	21.960	0.005
		9	0.002	0.041	21.960	0.009
		10	-0.081	-0.116	22.358	0.013
		11	-0.082	-0.052	22.778	0.019
		12	0.047	0.104	22.921	0.028
		13	-0.015	-0.281	22.935	0.042
		14	-0.102	-0.044	23.649	0.050
		15	-0.181	-0.103	25.955	0.038
		16	-0.185	-0.153	28.448	0.028
		17	-0.312	-0.202	35.784	0.005
		18	-0.258	-0.075	40.984	0.002
		19	-0.219	-0.208	44.876	0.001
		20	-0.092	0.147	45.592	0.001

Dependent Variable: INCENTIVO CONTRABANDO VI SIN TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 14:58
Sample: 1980 2024
Included observations: 45
Convergence achieved after 6 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.496312	0.138481	3.583962	0.0009
SIGMASQ	0.050511	0.008554	5.904878	0.0000

R-squared	0.245428	Mean dependent var	-5.43E-17
Adjusted R-squared	0.227879	S.D. dependent var	0.261650
S.E. of regression	0.229913	Akaike info criterion	-0.052520
Sum squared resid	2.272981	Schwarz criterion	0.027776
Log likelihood	3.181703	Hannan-Quinn criter.	-0.022587
Durbin-Watson stat	1.776493		

Inverted AR Roots	.50

Date: 11/05/26 Time: 14:58
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.098	0.098	0.4644	
		2	-0.142	-0.153	1.4494	0.229
		3	-0.172	-0.145	2.9332	0.231
		4	0.054	0.068	3.0826	0.379
		5	-0.032	-0.092	3.1353	0.535
		6	0.175	0.190	4.7875	0.442
		7	0.244	0.230	8.0987	0.231
		8	0.005	-0.020	8.1003	0.324
		9	-0.034	0.111	8.1674	0.417
		10	-0.075	-0.047	8.5092	0.484
		11	-0.084	-0.112	8.9504	0.537
		12	0.145	0.190	10.293	0.504
		13	-0.003	-0.209	10.294	0.590
		14	-0.049	-0.067	10.455	0.656
		15	-0.099	-0.043	11.141	0.675
		16	0.031	-0.091	11.212	0.737
		17	-0.222	-0.196	14.949	0.528
		18	-0.072	-0.068	15.358	0.570
		19	-0.139	-0.322	16.926	0.528
		20	-0.021	-0.025	16.964	0.592

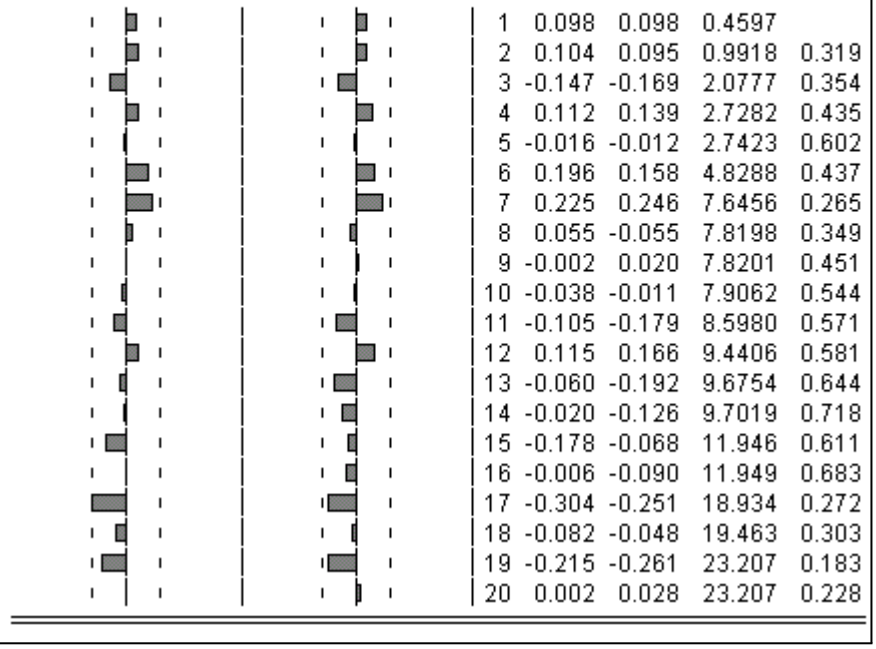
Dependent Variable: INCENTIVO_CONTRABANDO_VI_SIN_TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/02/26 Time: 13:20
Sample: 1980 2024
Included observations: 45
Convergence achieved after 42 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.478994	0.150264	3.187685	0.0027
SIGMASQ	0.050568	0.008774	5.763403	0.0000

R-squared	0.244574	Mean dependent var	-5.43E-17
Adjusted R-squared	0.227006	S.D. dependent var	0.261650
S.E. of regression	0.230043	Akaike info criterion	-0.051882
Sum squared resid	2.275551	Schwarz criterion	0.028414
Log likelihood	3.167355	Hannan-Quinn criter.	-0.021949
Durbin-Watson stat	1.771654		

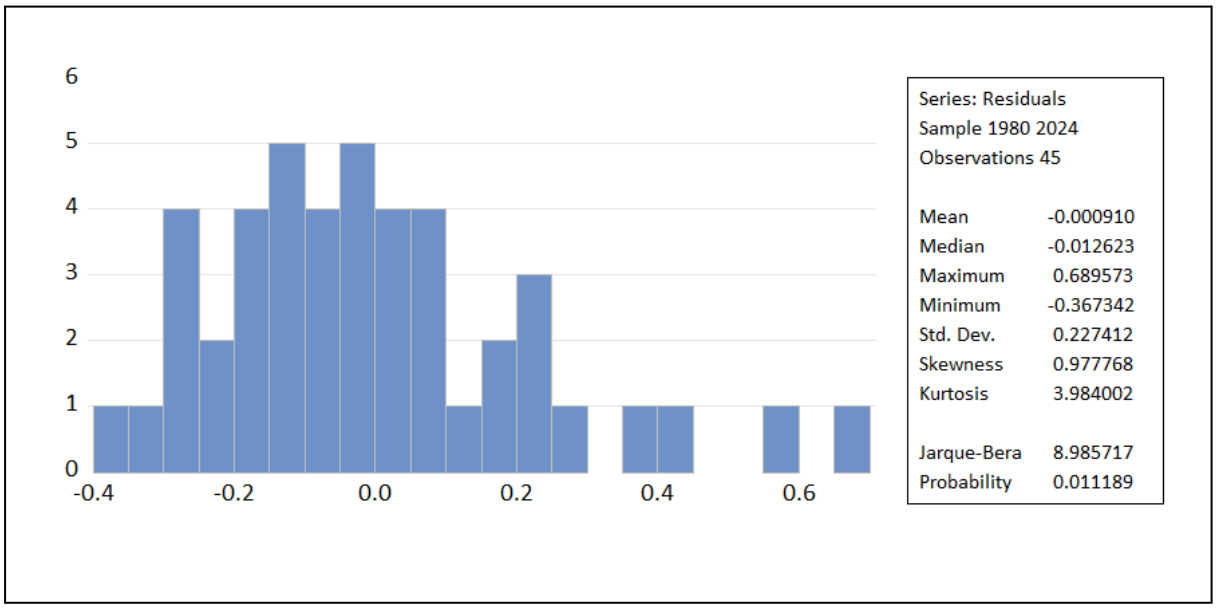
Inverted MA Roots	-.48

Date: 11/02/26 Time: 13:21
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.098	0.098	0.4597	
		2	0.104	0.095	0.9918	0.319
		3	-0.147	-0.169	2.0777	0.354
		4	0.112	0.139	2.7282	0.435
		5	-0.016	-0.012	2.7423	0.602
		6	0.196	0.158	4.8288	0.437
		7	0.225	0.246	7.6456	0.265
		8	0.055	-0.055	7.8198	0.349
		9	-0.002	0.020	7.8201	0.451
		10	-0.038	-0.011	7.9062	0.544
		11	-0.105	-0.179	8.5980	0.571
		12	0.115	0.166	9.4406	0.581
		13	-0.060	-0.192	9.6754	0.644
		14	-0.020	-0.126	9.7019	0.718
		15	-0.178	-0.068	11.946	0.611
		16	-0.006	-0.090	11.949	0.683
		17	-0.304	-0.251	18.934	0.272
		18	-0.082	-0.048	19.463	0.303
		19	-0.215	-0.261	23.207	0.183
		20	0.002	0.028	23.207	0.228

Comparo ambos modelos según el criterio de Akaike, en donde gana el modelo con AR(1) (-0.05252 < -0.05188), pero como la diferencia es tan pequeña, de tan solo, 0.00064, elijo el modelo MA(1) por presentar un mejor test de Jarque-Bera.



Al agregar esta primera variable de intervención se ve que se pudo obtener un modelo con residuos ruido blanco y coeficientes significativos, sin embargo sigue siendo problemático el hecho de que Dickey Fuller rechace la presencia de tendencia determinística y sin embargo resulta muy significativa en las estimaciones, a pesar de que analizando el gráfico solo se ve una tendencia en el periodo 1983-2000, lo que concuerda con las reformas económicas

de Ghana (ERP) ocurrida durante ese periodo, por lo tanto incorporo una nueva variable de intervención que refleje ese cambio de pendiente (SLS), la cual esta construida con valores igual a 0 para el periodo anterior a 1983, y luego tiene un crecimiento lineal (1, 2, 3, …) desde 1983 hasta el año 2000, y valores iguales a los de el año 2000 (18) hasta finalizar la serie.
Además la variable de intervención para el año 2004, dejó de ser una variable aditiva de única intervención, sino que pasó a ser de cambio transitorio (TC), que tiene su pico máximo en 2004, y su impacto va decayendo año a año a un nivel gamma (δ) de 0.5.

Dependent Variable: INCENTIVO CONTRABANDO
Method: Least Squares
Date: 11/05/26 Time: 11:55
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
TC 2004	1.511684	0.179457	8.423677	0.0000
SLS IC	0.037937	0.009592	3.954919	0.0003
@TREND	-0.000742	0.004598	-0.161420	0.8726
C	0.332188	0.067687	4.907727	0.0000

R-squared	0.807315	Mean dependent var	0.891044
Adjusted R-squared	0.793216	S.D. dependent var	0.421261
S.E. of regression	0.191562	Akaike info criterion	-0.382520
Sum squared resid	1.504540	Schwarz criterion	-0.221928
Log likelihood	12.60670	Hannan-Quinn criter.	-0.322653
F-statistic	57.26068	Durbin-Watson stat	1.665702
Prob(F-statistic)	0.000000		

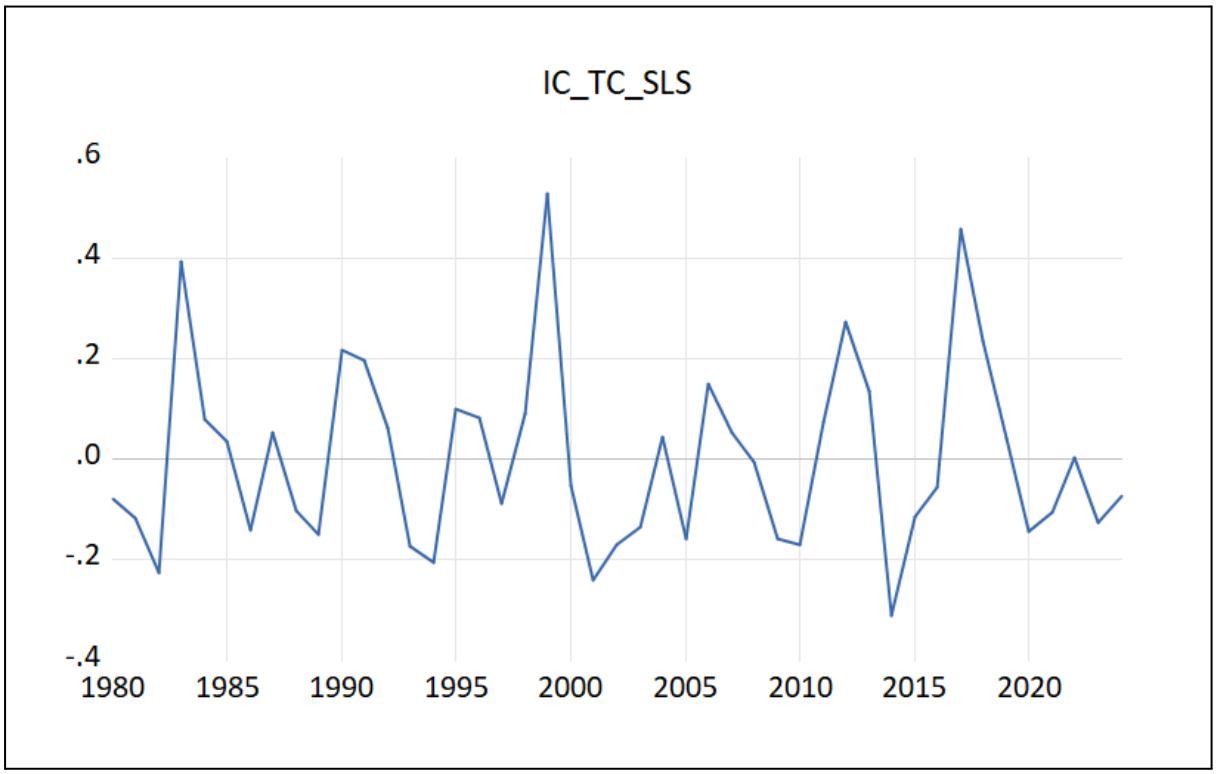
De este modo la tendencia determinística deja de ser significativa, por lo que no se encuentra una tendencia lineal que abarque todo el periodo 1980-2024, ya que la variable de intervención absorbió ese impacto.

Dependent Variable: INCENTIVO_CONTRABANDO
Method: Least Squares
Date: 11/04/26 Time: 22:43
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
TC_2004	1.518167	0.172864	8.782426	0.0000
C	0.333785	0.066179	5.043689	0.0000
SLS_IC	0.036578	0.004542	8.053344	0.0000

R-squared	0.807192	Mean dependent var	0.891044
Adjusted R-squared	0.798011	S.D. dependent var	0.421261
S.E. of regression	0.189328	Akaike info criterion	-0.426329
Sum squared resid	1.505497	Schwarz criterion	-0.305885
Log likelihood	12.59241	Hannan-Quinn criter.	-0.381429
F-statistic	87.91671	Durbin-Watson stat	1.662997
Prob(F-statistic)	0.000000		

A simple vista se ve como tanto la varianza como el AIC mejoró con respecto a los modelos anteriores sin variable de intervención y con una única variable de intervención aditiva.



Null Hypothesis: IC_TC_SLS has a unit root
Exogenous: None
Lag Length: 1 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-6.093932	0.0000
Test critical values:	1% level	-2.619851	
	5% level	-1.948686	
	10% level	-1.612036	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(IC_TC_SLS)
Method: Least Squares
Date: 11/04/26 Time: 22:44
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
IC_TC_SLS(-1)	-1.147134	0.188242	-6.093932	0.0000
D(IC_TC_SLS(-1))	0.364987	0.145463	2.509134	0.0161

R-squared	0.498541	Mean dependent var	0.000975
Adjusted R-squared	0.486311	S.D. dependent var	0.244087
S.E. of regression	0.174942	Akaike info criterion	-0.603329
Sum squared resid	1.254794	Schwarz criterion	-0.521412
Log likelihood	14.97157	Hannan-Quinn criter.	-0.573121
Durbin-Watson stat	2.236215		

Date: 11/04/26 Time: 22:44
Sample: 1980 2024
Included observations: 45

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.165	0.165	1.3011	0.254
		2	-0.325	-0.362	6.5086	0.039
		3	-0.407	-0.324	14.838	0.002
		4	-0.034	-0.041	14.896	0.005
		5	0.065	-0.200	15.123	0.010
		6	-0.035	-0.251	15.188	0.019
		7	0.144	0.132	16.339	0.022
		8	0.121	-0.045	17.175	0.028
		9	0.035	0.002	17.248	0.045
		10	-0.281	-0.201	22.023	0.015
		11	-0.136	-0.046	23.174	0.017
		12	0.073	-0.061	23.516	0.024
		13	0.129	-0.101	24.620	0.026
		14	0.033	-0.069	24.695	0.038
		15	-0.076	-0.117	25.105	0.049
		16	0.067	0.006	25.433	0.063
		17	-0.138	-0.281	26.863	0.060
		18	-0.039	-0.032	26.985	0.079
		19	0.011	-0.091	26.995	0.105
		20	0.060	-0.271	27.294	0.127

Dependent Variable: IC_TC_SLS
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/04/26 Time: 22:52
Sample: 1980 2024
Included observations: 45
Convergence achieved after 12 iterations
Coefficient covariance computed using outer product of gradients

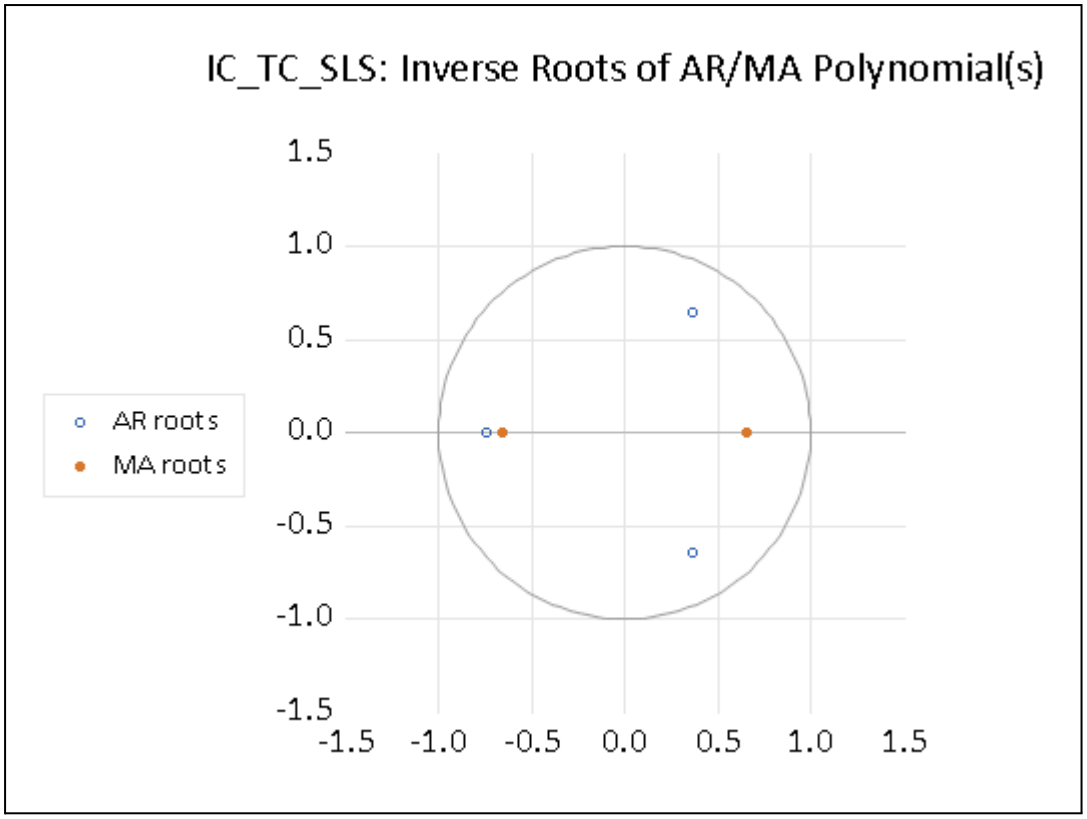
Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(3)	-0.406601	0.189518	-2.145447	0.0377
MA(2)	-0.423696	0.205844	-2.058337	0.0458
SIGMASQ	0.023659	0.003976	5.950293	0.0000

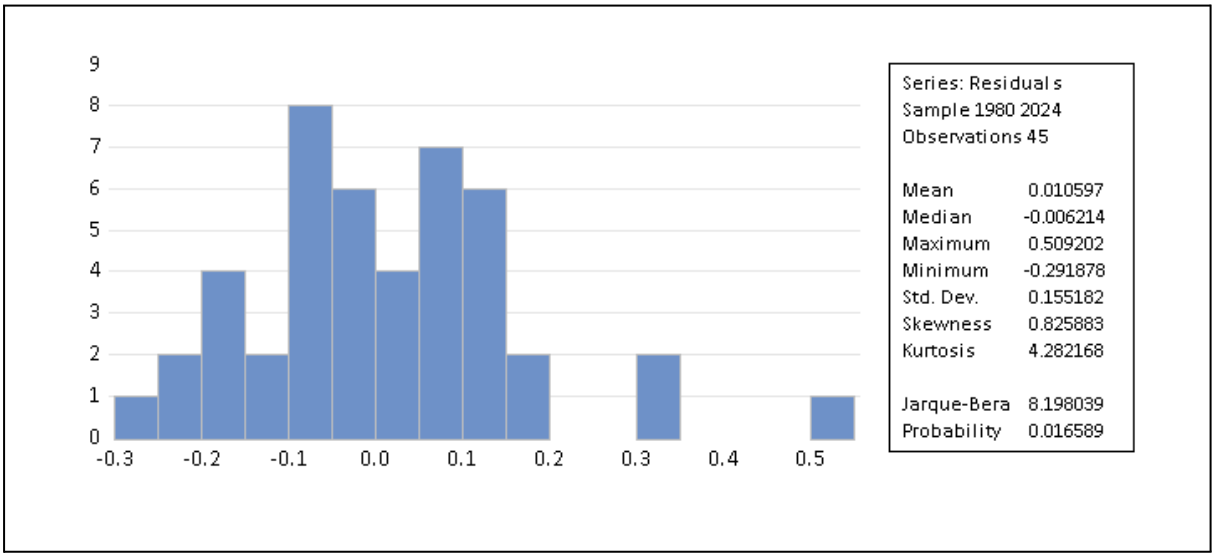
R-squared	0.292832	Mean dependent var	-8.64E-17
Adjusted R-squared	0.259158	S.D. dependent var	0.184975
S.E. of regression	0.159212	Akaike info criterion	-0.752538
Sum squared resid	1.064638	Schwarz criterion	-0.632094
Log likelihood	19.93210	Hannan-Quinn criter.	-0.707637
Durbin-Watson stat	1.962562		

Inverted AR Roots	.37-.64i	.37+.64i	-.74
Inverted MA Roots	.65	-.65	

Date: 11/04/26 Time: 22:54
Sample: 1980 2024
Q-statistic probabilities adjusted for 2 ARMA terms

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.006	0.006	0.0019	
		2	0.001	0.001	0.0019	
		3	-0.122	-0.122	0.7455	0.388
		4	0.018	0.020	0.7625	0.683
		5	-0.020	-0.021	0.7842	0.853
		6	-0.228	-0.246	3.5978	0.463
		7	0.083	0.099	3.9854	0.552
		8	-0.003	-0.011	3.9859	0.679
		9	0.100	0.039	4.5679	0.713
		10	-0.266	-0.255	8.8574	0.354
		11	-0.102	-0.118	9.5017	0.392
		12	-0.042	-0.082	9.6146	0.475
		13	-0.004	-0.039	9.6157	0.565
		14	-0.021	-0.066	9.6465	0.647
		15	-0.157	-0.187	11.386	0.579
		16	0.169	0.023	13.464	0.490
		17	-0.175	-0.267	15.787	0.396
		18	0.057	-0.023	16.040	0.450
		19	0.073	0.115	16.475	0.490
		20	0.023	-0.167	16.520	0.556





Este modelo luego de aplicar las variables de intervención, resulta un AR(3) MA(2) con ambos coeficientes significativos al 5% utilizando el modelo de estimación de Máxima Verosimilitud (ML), ,con residuos ruido blanco, y raíces invertidas dentro del círculo unitario, y con un criterio de Akaike de -0.75, muy inferior al del modelo anterior con variable de intervención (-0.70 < 0.03).